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Acute impact of a psychedelic injection on neural oscillations in the mouse insular cortex

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Abstract

Anxiety disorders are among the most prevalent psychiatric conditions, yet current treatments remain only partially effective. Psychedelic compounds have re-emerged as promising tools for both clinical intervention and neuroscientific discovery, showing rapid and sustained anxiolytic effects in both humans and animal models. However, the neural mechanisms underlying these effects—especially at the level of circuit dynamics—remain incompletely understood.

Here, we investigated the acute impact of 5-MeO-DMT, a potent fast-acting tryptamine, on neuronal dynamics in the anterior insular cortex (aIC) of freely behaving mice using local field potential (LFP) recordings and single-unit electrophysiology. Spectral analyses revealed a paradoxical signature: despite a widespread suppression of neuronal firing, we observed a robust increase in total spectral power, a steepening of the aperiodic spectral slope, and the emergence of gamma-band oscillations (~60 Hz), accompanied by a collapse of alpha rhythms (~8–11 Hz).

These findings suggest a decoupling between spiking output and spectral synchrony, pointing to a complex reorganization of excitation/inhibition (E/I) balance under psychedelics. Enhanced inhibitory tone may coexist with rhythmic coordination mediated by interneuron-dominated microcircuits. The results align with human studies reporting decreased alpha, increased gamma, and elevated brain entropy under psychedelics, and highlight the aIC as a key hub linking interoception, emotional salience, and behavioral expression—particularly immobility and backward locomotion.

Methodologically, this study combines multiscale electrophysiological analysis (PSD, FOOOF, spike sorting), waveform-based classification, and behavioral quantification.

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INTRODUCTION

1. Scientific Background

Anxiety is a natural, evolutionary conserved response present in anticipation of a danger, which is essential for survival and adaptation across species (American Psychiatric Association, 2013). However, when it becomes chronic, disproportionate, or disconnected from actual environmental cues, it constitutes pathological anxiety disorders—among the most prevalent psychiatric conditions worldwide. Globally, anxiety disorders affect approximately 18% of the adult population annually and up to 28% over a lifetime (Yang et al., 2021). Clinically, anxiety can be conceptualized in two forms: a stable, trait-like predisposition (trait anxiety), and a transient, state-dependent response to a perceived threat (Spielberger, 1983; Endler & Kocovski, 2001; Daviu et al., 2019).

Due to the partial efficacy of current treatments—such as selective serotonin reuptake inhibitors (SSRIs) and cognitive-behavioral therapy—new avenues for both therapeutic intervention and neuroscientific inquiry are actively being pursued (Garakani et al., 2006; Bandelow et al., 2017). Among these, psychedelic compounds have emerged as powerful tools, not only for modulating conscious states, but also for probing the deeper functional organization of the brain. Substances such as psilocybin, DMT, and the methoxylated tryptamine (5-MeO-DMT) have recently garnered renewed interest in neuroscience and psychiatry due to their rapid and durable effects on mood and anxiety disorders. Studies in humans revealed that a single administration of either 5-MeO-DMT or psilocybin produces a significant reduction in anxiety symptoms five weeks post-treatment. Likewise, experiments in mice have shown that injection of these same compounds leads to decrease in anxiety-like behaviors (Davis et al., 2019; Griffiths et al., 2016).

Within this neurobiological framework, the anterior insular cortex (aIC) occupies a particularly strategic position. Located at the interface of sensory, affective, and cognitive systems, the aIC is considered a hub for interoceptive integration, emotional valence processing, and self-awareness (Critchley et al., 2004; Paulus & Stein, 2006; Mathew et al., 2008). Interestingly, meta-analyses of anxious patients have revealed hyperactivation of the aIC in response to negative stimuli, suggesting dysregulation of this region in anxiety (Etkin & Wager, 2007; Klumpp et al., 2012; Shah et al., 2009; Gehrlach et al., 2019). In mice, recent studies have shown that activation of glutamatergic neurons in the aIC promotes anxiogenic-like behavior, while inhibition of the same circuit produces anxiolytic-like effects—especially via projections to the basolateral amygdala (Nicolas et al., 2023).

Understanding how aIC neuronal activity is reconfigured under psychedelics may therefore provide novel insights into the neural underpinnings of anxiety—and open new avenues for therapeutic modulation of emotional brain states.

Animal models are essential for elucidating the neural mechanisms behind these effects. In particular, rodents offer the advantage of controlled pharmacological interventions alongside detailed behavioral and neuronal measurements, making them a powerful translational model for investigating psychedelic action

2. Behavioural Signatures of Psychedelic States in Mice

Psychedelic compounds induce distinct behavioural, cognitive, and neural signature in mice, markedly distinct from conventional psychopharmacological responses. Unlike stimulants or anxiolytics such as benzodiazepines, serotonergic psychedelics like 5-MeO-DMT induce behavioural states that fluctuate between motor suppression, bursts of agitation, and postural disruptions. In mice, this typically manifests as abrupt immobility, tremors, darting, body flattening, and lateral splay.

Two behavioral markers are particularly informative in characterizing this state: the head-twitch response and the locomotion dynamics. Three behavioural markers are particularly informative in characterizing this state. The head-twitch response (HTR)—a rapid, stereotyped head jerk—serves as a primary indicator of 5-HT_{2A} receptor activation and hallucinogenic action. During the HTR, torsional rotations of the head and neck can propagate through the body as full-body shake,

Locomotion dynamics impairment, may reflect disruptions in sensorimotor integration or altered egocentric spatial mapping... In particular , **backward locomotion**, an ethologically atypical behaviour that consistently emerges under psychedelic conditions and may reflect altered sensorimotor coordination or egocentric mapping.

3. Neural Oscillations: Quantifying the Rhythm of the Brain

Neural oscillations—also known as brain rhythms—are a fundamental feature of collective neuronal activity. They arise from the synchronized interplay between excitatory and inhibitory neuronal populations and are observable in local field potentials (LFPs), EEG, and MEG recordings (Buzsáki, 2006). LFPs, in particular, reflect low-frequency extracellular voltage fluctuations that capture the summed synaptic activity of nearby neurons. These rhythmic patterns reflect temporally structured fluctuations in extracellular voltage, generated by coordinated post-synaptic currents across neuronal ensembles.

Rather than constituting background noise, oscillations are now recognized as essential for coordinating neural communication. Distinct frequency bands have been associated with specific cognitive and behavioral states: delta (1–3 Hz) and theta (4–7 Hz) dominate during sleep and navigation, alpha (8–12 Hz) is linked to quiet wakefulness and inhibitory control, beta (13–30 Hz) to motor processing, and gamma (30–100 Hz) to attention, sensory integration, and conscious awareness.

Critically, alterations in oscillatory dynamics are observed across several neuropsychiatric disorders including anxiety, schizophrenia, and depression—highlighting their relevance for brain function. Psychedelics offer a unique opportunity to probe these dynamics by transiently disrupting large-scale brain organization. Studies have shown that serotonergic psychedelics such as DMT and psilocybin decrease alpha-band power—particularly in the default-mode network—and concurrently increase gamma-band oscillations, which are thought to reflect heightened excitability and altered sensory integration (Muthukumaraswamy et al., 2013; Timmermann et al., 2023).

These effects are accompanied by a shift in the excitation/inhibition (E/I) balance within cortico-thalamic and cortico-cortical circuits, and by a marked increase in total brain energy and spectral power, especially in higher frequency ranges (Atasoy et al., 2018). Altogether, these features suggest that psychedelics disorganize canonical rhythms while promoting novel modes of neural communication, thus offering a powerful model to investigate the role of oscillations in consciousness and psychiatric dysfunction.

3.2 Spectral decomposition: capturing frequency-specific activity

In order to quantify these rhythmic components, electrophysiological signals must be decomposed from the time domain into the frequency domain. This process—known as **spectral decomposition**—allows us to determine **how much power is contained at each frequency**.

This is done by applying a mathematical transformation to the signal, typically a **Fourier Transform**, which breaks the signal into its constituent sine waves. Because real-world signals like LFPs are not stationary (they change over time), we often use **Short-Time Fourier Transform (STFT)** to obtain a **time–frequency representation**. This produces a **spectrogram**, showing how frequency power evolves over time. While spectral decomposition provides a dynamic view of how oscillatory power fluctuates over time, it is often complemented by more global, time-averaged measures that summarize frequency content across longer epochs. Among these, the Power Spectral Density (PSD) stands out as a fundamental tool for characterizing the overall structure of neural signals in the frequency domain.

3.3 Power Spectral Density (PSD): definition and interpretation

The **Power Spectral Density (PSD)** is a widely used metric that quantifies the distribution of signal power across frequencies. In essence, it tells us **how much variance (or energy)** in the signal is attributable to each frequency component.

The PSD provides rich information:

- It allows identification of **dominant frequency bands** (e.g., increased gamma power),
- It helps detect global shifts in spectral energy, and
- It enables comparisons across experimental conditions (e.g., pre- vs. post-injection).

However, the PSD has **key limitations**. The spectrum of LFPs follows a **1/f-like decay**, meaning power tends to decrease with frequency. This aperiodic background can obscure or distort the interpretation of true oscillatory peaks. In addition, simple band-averaging may miss subtle shifts in frequency or changes in spectral shape that carry functional significance.

3.4 Parametric modeling with FOOOF: isolating meaningful peaks

To address these limitations, **Fitting Oscillations and One Over F (FOOOF)** model has been developed (Donoghue et al., 2020). FOOOF is a **data-driven algorithm** that decomposes the power spectrum into two components:

- The **aperiodic component** (also called “background” or “1/f”), which follows a power-law decay,
- And the **periodic components**, modeled as Gaussian peaks that reflect putative oscillations.

The aperiodic part is characterized by two parameters:

- The **slope**, which describes how fast the power decreases with frequency,
- The **offset**, which reflects the overall vertical shift of the spectrum (baseline power).

Oscillatory peaks are identified by fitting Gaussians distributionson top of this background. FOOOF thus allows us to distinguish genuine frequency-specific oscillations from changes in broadband activity—something traditional PSD methods cannot achieve.

Together, PSD and FOOOF provide a **complementary view** of oscillatory brain dynamics—one focusing on spectral content, the other on spectral structure. Their joint application is essential to uncover nuanced neurophysiological effects

Importantly, recent work has shown that the aperiodic slope and offset themselves are biologically meaningful: the slope may reflect excitatory/inhibitory balance, and the offset may relate to overall neuronal spiking or firing rate (Donoghue et al., 2020; Waschke et al., 2021). This interpretation suggests that spectral features extracted from LFPs may carry indirect signatures of underlying spiking activity. However, to directly probe how psychedelics reshape neuronal output at the cellular level, we must turn to single-unit electrophysiology.

4. Single-Unit Activity: Exploring the Microcircuitry of the Insula

To fully characterize neural dynamics in the aIC, it is not sufficient to examine oscillatory activity alone. While LFPs reveal coordinated population-level fluctuations, they do not capture the

contributions of individual neurons—nor do they resolve the heterogeneity of neural responses to pharmacological perturbation.

This is where **single-unit electrophysiology** becomes essential: By applying a bandpass filter to the extracellular signal, high-frequency voltage deflections caused by action potentials can be isolated from slower components like LFPs. These fast events, reflecting the activity of multiple nearby neurons—referred to as multi-unit activity (MUA)—are then disentangled through spike sorting, a process that clusters similar waveforms into putative single units. This approach allows precise monitoring of individual firing patterns over time, offering high-resolution insight into the microcircuit-level effects of neuromodulators like psychedelics, particularly within the aIC, where diverse neuronal populations interact dynamically.

4.1 Classifying putative excitatory and inhibitory neurons

Beyond isolating units, an important goal is to infer their **functional identity**. While intracellular labeling is required for definitive classification, several extracellular features have been statistically associated with excitatory or inhibitory types. For instance, inhibitory interneurons often exhibit **narrow spike waveforms**, **shorter peak-to-trough durations**, and **higher firing rates** than excitatory pyramidal cells (Barthó et al., 2004; Trainito et al., 2019).

4.2 Quantifying response profiles and functional modulation

To understand how single units were affected by 5-MeO-DMT, we examined changes in their firing rates relative to the moment of injection. Each unit’s activity was compared to its own pre-injection baseline, allowing us to determine whether it showed a statistically meaningful increase or decrease in spiking—classified respectively as **excited**, **inhibited**, or **unchanged**.

5. Aim of the Study

In this study, we investigated the acute effects of 5-MeO-DMT—a fast-acting tryptamine psychedelic—on neuronal activity in the anterior insular cortex of freely behaving mice. To this end, we combined high-density in vivo electrophysiological recordings with behavioral measurements in semi-naturalistic conditions.

Specifically, we examined:

- the **oscillatory dynamics** of local field potentials (LFPs) using both spectral and parametric approaches (e.g., FOOOF),
- the **modulation of single-unit activity** via spike sorting and waveform-based classification,

and the study of behavioral markers of psychedelic action, specifically head-twitch responses and alterations in locomotion such as immobility and backward walking. Our goal was **to describe how 5-MeO-DMT alters oscillatory and unit-level activity in the aIC**, both in terms of global activity and fine-grained neuronal organization;

MATERIAL AND METHODS

All experimental procedures were approved by the local ethics committee of the University of Bordeaux and authorized by the French Ministry of Research (MESR), in compliance with European Union guidelines for animal research (Directive 2010/63/EU).

1. Experimental Design

Two distinct cohorts of C57BL/6 mice were used to investigate the acute effects of the serotonergic psychedelic 5-MeO-DMT (20 mg/kg, intraperitoneally) on spontaneous behavior and neural activity. All experiments were conducted under a unified protocol, consisting of a 10-minute pre-injection baseline followed by 60 minutes of continuous observation in a standardized Open Field Test (OFT) arena. Environmental conditions—including lighting, sound, and arena dimensions—were kept constant across all sessions.

- **Behavioral cohort:** 31 non-implanted mice were used to characterize the behavioral effects of 5-MeO-DMT. Mice were randomly assigned to one of two groups: 5-MeO-DMT ($n = 17$; 8 males, 9 females) or saline ($n = 14$; 7 males, 7 females).
- **Electrophysiological cohort:** 11 mice were chronically implanted with a 16-channel microelectrode array targeting the right anterior insular cortex (aIC). These mice were also split into two groups: 5-MeO-DMT ($n = 6$; 3 males, 3 females) and saline ($n = 5$; 3 males, 2 females).

All procedures adhered to institutional and national ethical guidelines for animal experimentation.

2. Stereotaxic Surgery and Electrode Implantation

Implantation of chronic electrodes was performed under deep anesthesia using stereotaxic guidance on 10 weeks of age mice. The anterior insular cortex (right hemisphere) was targeted using the following coordinates relative to Bregma: anterior–posterior (AP) +1.7 mm, medio-lateral (ML) +3.1 mm, and dorsal–ventral (DV) –3.5 mm. A 4×4 tungsten microelectrode array (Innovative Neurophysiology; 35 μm wire diameter, 150 μm inter-electrode spacing) was implanted vertically into the cortex. A silver ground wire (diameter 200 μm) was inserted at a distant site for signal referencing. After surgery, animals were allowed to recover for a minimum of seven days before recording, and monitored daily for general health status.

3. Behavioral analysis

3.1. Head-twitches

The head-twitch response (HTR) are a classical marker of psychedelic action via 5-HT_{2A} receptor activation are defined as a rapid and stereotyped dorsoventral jerk of the head. they were observed in both groups but occurred at a markedly higher frequency in the 5-MeO-DMT-treated mice. HTR

were detected using an electromagnetic setup: a small magnet was affixed to the mouse's ear, and the animal was placed inside a coil of conductive copper wire. Each head-twitch generated a transient signal amplified and recorded continuously for 60 minutes post-injection. HTR events were automatically detected from the signal and later visually confirmed using synchronized video recordings (Figure 1C)

3.2. Locomotion – Figure 1.D

3.2.1 Video acquisition and coordinate extraction

Spontaneous behavior was recorded continuously for 70 minutes at 30 frames per second using an overhead video camera positioned above the open-field arena. Video recordings were analyzed using **DeepLabCut (DLC)**, a supervised deep learning framework for pose estimation, retrained on our dataset. Seven anatomical keypoints were labeled along the mouse's longitudinal axis: Tail_base, Center, Spine_1, Spine_2, Spine_3, Neck, and Nose. For each frame, the 2D pixel coordinates of these points were extracted and later converted into physical units (centimeters) using a session-specific pixel-to-cm conversion factor, calibrated via OpenCV.

3.2.2 Movement direction detection via PCA.

To determine movement direction (forward vs. backward), we used a method based on global body orientation derived from Principal Component Analysis (PCA). For each frame, a direction vector was computed using PCA on the coordinates of three rigid points along the spine: *Tail_base*, *Center*, and *Spine_1*.

However, PCA alone does not guarantee anatomically correct orientation (i.e., tail-to-head), since it does not capture polarity. To resolve this, a reference vector was calculated between the anatomical endpoints of the body—*Tail_base* → *Spine_1*—and considered to point forward. The dot product between this reference vector and the first principal component was then computed. If the product was negative, this indicated that the PCA vector was reversed with respect to the real anatomical axis; it was then flipped (multiplied by -1) to ensure directional consistency. This step ensures that all body direction vectors point toward the animal's front, which is crucial for reliably detecting *backward* movements.

This approach was preferred over classical linear regression because it does not depend on the ordering of points and captures the principal direction of variance in the body shape, making it more robust to posture variations and tracking errors. The first principal component (PC1) provides a unit vector representing the animal's longitudinal axis.

3.2.3 Projected displacement detection.

For each pair of frames spaced 15 frames apart (0.5 s), the position of the mouse's center (*Center*) at time t and $t+15$ was computed. This displacement vector was then orthogonally projected onto the body direction vector extracted at time t using a dot product. The result of this projection (a positive or negative scalar value) measures the displacement component along the body's longitudinal axis.

Thus, it is not merely an angle between movement and orientation vectors, but a **direct projection** of motion along the body axis—offering greater sensitivity to subtle movements, particularly backward motion.

3.2.4 Decision thresholds and categorization.

To discriminate between movement types, a threshold was defined using a morphological metric: the average distance between anterior and posterior hips, measured automatically via orthogonal projection. This threshold (in pixels) served as a criterion to detect meaningful movement.

Three movement categories were defined:

- **Forward:** positive projection exceeding the forward threshold.
- **Backward:** negative projection below the backward threshold.
- **Immobile:** projected displacement within an intermediate (noise/micromovement) range.

A special case was added to detect **turning**, triggered when the angle between body direction vectors at t and $t+15$ exceeded 65° , indicating a major change in orientation.

3.2.5 Post hoc correction.

To reduce misclassification due to isolated frames, label smoothing was applied: sequences of fewer than three consecutive frames classified as "immobile" but surrounded by a consistent linear movement type ("forward" or "backward") were retrospectively reclassified.

4. Electrophysiological Recordings

Recordings were performed using the Open Ephys acquisition system with a sampling rate of 30 kHz. Signals were referenced online using a common average reference (CAR) to suppress shared noise artifacts. Both local field potentials (LFPs) and spiking activity were acquired continuously for the full 70-minute session (10 minutes baseline, 60 minutes post-injection).

The quality of electrophysiological recordings was verified for each channel. Channels showing flat signals (standard deviation $< 5 \mu\text{V}$) or excessive noise (standard deviation $> 500 \mu\text{V}$) were excluded. Channel 5 was systematically selected for LFP analyses across all mice based on both signal stability and stereotaxic considerations, as it corresponded to a central electrode in the 4×4 matrix.

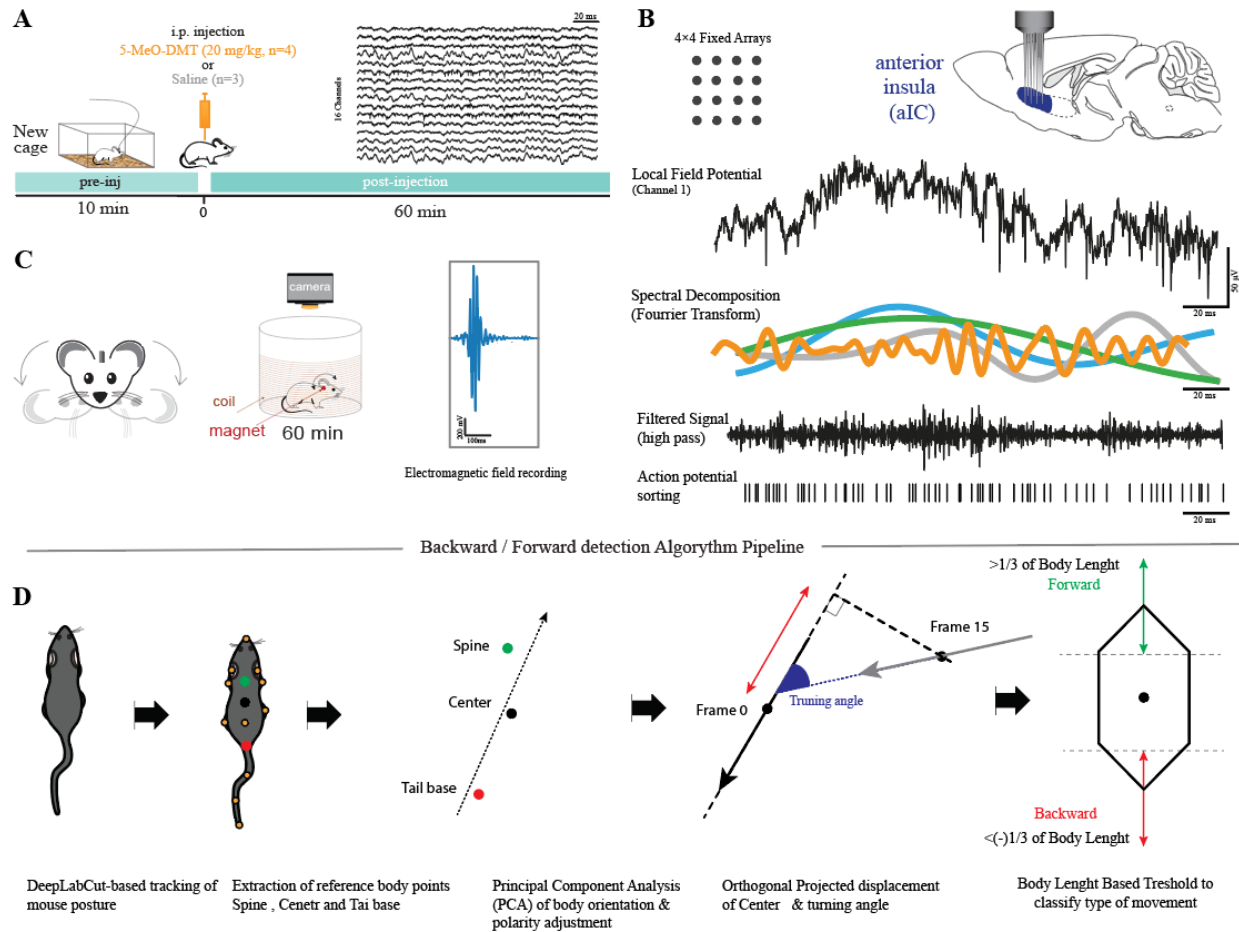


Figure 1. Experimental design & data acquisition.

1.A. Experimental timeline and representative LFP recording. Mice were placed in a new cage for baseline (pre-injection) electrophysiological recordings lasting 10 minutes, followed by intraperitoneal injection (i.p.) of either 5-MeO-DMT (20 mg/kg, $n = 4$) or saline ($n = 3$). Neural activity was then continuously recorded for 60 minutes post-injection. Right panel: example raw LFP traces from 30 channels over a 20-second period; vertical and horizontal scale bars denote 20 μ V and 20 s, respectively. **1.B. Electrophysiological recordings and signal processing in the anterior insular cortex** were performed using fixed 4x4 microelectrode arrays targeting the anterior insular cortex (aIC). Extracted signals included local field potentials (LFP), bandpass-filtered signals, and sorted single-unit spikes. Fourier-based spectral decomposition was applied to LFP traces for time-frequency analysis. **1.C. Electromagnetic detection of head-twitch response (HTR).** HTRs were detected by placing a magnet on the mouse's ear and recording electromagnetic transients from a surrounding copper coil. Signals were amplified and recorded for 60 minutes post-injection, with automatic detection confirmed by video. **1.D. Behavioral tracking and movement direction classification pipeline.** Motion tracking and direction classification pipeline. DeepLabCut was used to track mouse posture (spine, center, tail base). Principal Component Analysis (PCA) was applied to estimate body orientation and polarity. Frame-to-frame displacements of the center point, projected orthogonally to the body axis, were used to compute movement direction. Movements were classified as forward or backward based on a body length-normalized threshold ($>1/3$ for forward, $<1/3$ for backward).

4.1. Spectral analysis

4.1.2 Power Spectral Density (PSD)

4.1.2.1 Spectral Decomposition of LFP Signals

To examine the impact of 5-MeO-DMT on cortical oscillations, we first transformed the raw local field potential (LFP) data into time–frequency representations using a **Short-Time Fourier Transform (STFT)** Fig.3A, implemented via the `scipy.signal.spectrogram` function. The parameters were carefully selected to optimize both temporal and frequency resolution with a **Window length** of 1.09 seconds and **50% overlap with a** : 48–52 Hz **Exclusion zone** remove 50 Hz line noise artifacts (Notch Filter)

This process yielded a spectrogram $S(f, t)$ for each mouse, capturing the instantaneous power at each frequency f and time point t

To allow for comparisons across mice and timepoints, the spectrogram was **normalized to a pre-injection baseline**, defined as the “*before_injection*” interval:

$$S_{\text{norm}}(f, t) = 10 \cdot \log_{10} \left(\frac{S(f, t)}{P_{\text{baseline}}(f)} \right)$$

Where $S(f, t)$ is the raw power at frequency f and time t , $P_{\text{baseline}}(f)$ is the mean spectral power across all time points in the baseline window.

This normalization in decibels allows the detection of relative increases or decreases in spectral content post-injection. Each spectrogram was **time-aligned to the injection** and smoothed using a **2D Gaussian kernel** with standard deviation.

Group-level median spectrograms were computed across all mice within each condition (5-MeO-DMT vs. Saline).

4.1.2.2 Temporal Evolution of Canonical Bands

To better understand how different frequency bands evolve over time, we extracted time series of normalized power for five standard bands: delta (1–3 Hz), theta (3–8 Hz), alpha (8–11 Hz), beta (11–30 Hz), and gamma (30–80 Hz).

For each band b , power at time t was computed as:

$$P_b(t) = \frac{1}{|F_b|} \sum_{f \in F_b} S_{\text{norm}}(f, t)$$

These trajectories were smoothed using a temporal Gaussian filter and aggregated across mice. Group-level medians $\pm$ interquartile ranges are shown in Figure 3B (Saline) and Figure 3C (5-MeO-DMT), focusing on gamma, alpha, and delta bands

4.1.3 Parametric Modelisation : Fitting Oscillation and One Over F (FOOOF)

4.1.3.1 Introduction and Theoretical Rationale

To deepen our understanding of how 5-MeO-DMT modulates brain oscillations, we applied the FOOOF algorithm—**Fitting Oscillations and One-Over-F** (Donoghue et al., 2020). Unlike classical power spectral density (PSD) analyses that conflate oscillatory peaks and broadband background activity, FOOOF disentangles the **aperiodic components** (slope and offset of the 1/f background) from **periodic components** (true oscillatory peaks). This distinction is particularly important under psychedelics, where widespread excitability changes may globally shift power without implying functional rhythmic organization.

This method models the logarithm of the power spectrum as a sum of:

$$\log_{10} P(f) = b - \chi \cdot \log_{10}(f) + \sum_i A_i \cdot e^{\left(-\frac{(f-f_i)^2}{2w_i^2}\right)}$$

Where b is the **offset** (baseline power level), χ is the **slope** of the aperiodic background, and f_i, A_i , and w_i define respectively the **center frequency**, **amplitude**, and **bandwidth** of each oscillatory peak.

Biophysical interpretations of FOOOF parameters:

The **slope** is typically interpreted as a proxy for the **excitation/inhibition (E/I) balance**: a **steeper slope** (more negative) may reflect increased inhibition or reduced spiking, while a **flatter slope** (less negative) indicates heightened excitation or disinhibition.

The **offset** relates to the **overall amplitude of neural activity**—higher offset values suggest stronger global signal, potentially due to increased firing, neuromodulation, or synchrony.

Peaks identify structured rhythmic components (e.g., alpha, gamma), offering insight into **local network synchronization**.

4.1.3.2 Preprocessing and Spectral Estimation

LFP data were first notch-filtered at 50 Hz ($Q=50$) to remove line noise. No bandpass filtering or downsampling was applied, preserving full spectral content between 1 and 90 Hz.

Power spectra were computed using Welch's method (`nperseg = 16384`, `noverlap = 8192`), yielding a frequency resolution of approximately 1.83 Hz. The resulting spectra were log-transformed and restricted to the [1–90 Hz] range before FOOOF fitting.

The quality of each fit was visually inspected. We extracted the aperiodic parameters (slope, offset) and periodic features (peak frequency, peak power) for each spectrum.

4.2. Single unit analysis

Extracellular recordings capture both low-frequency local field potentials (LFPs) and high-frequency voltage deflections associated with action potentials. To isolate spiking activity, raw signals were first bandpass filtered (300–6000 Hz), allowing the extraction of high-frequency events generated by multiple nearby neurons—referred to as multi-unit activity (MUA).

Spike detection and sorting were performed using standard pipelines (e.g., **Kilosort**; Pachitariu et al., 2016), which identify negative transients crossing a threshold and cluster them based on waveform similarity. These automatically sorted spikes were then manually curated in **Phy2** (Rossant et al., 2016) to isolate high-quality single units. Only units labeled as “good” were retained for analysis, based on three main criteria, adapted from established practices (Barthó et al., 2004):

- **Waveform shape** (i.e., clear and stereotyped spike morphology),
- **Inter-spike interval (ISI) histogram** (showing a physiologically plausible firing pattern and minimal contamination from other units),
- **Absence of refractory period violations** (i.e., no spikes within the expected minimum inter-spike interval, typically <2 ms).

This procedure yielded a curated population of well-isolated neurons suitable for downstream classification.

4.2.1 Waveform-Based Clustering of Putative Excitatory and Inhibitory Neurons

To distinguish putative excitatory and inhibitory units within this population, two unsupervised clustering approaches were implemented using spike waveform features. In the absence of direct ground-truth labeling, this classification relied on morphological proxies—such as spike duration, symmetry, and repolarization dynamics—that have been consistently associated with cell type identity in extracellular recordings (Barthó et al., 2004; Trainito et al., 2019; Vaziri et al., 2019; Oghazian et al., 2021).

4.2.1.1 Waveform preprocessing

All units manually labeled as “good” in Phy2 were included. For each, we extracted the mean waveform from the main channel of detection (i.e., highest amplitude). The waveform was interpolated (cubic spline) for increased temporal resolution and aligned to its global minimum (the trough). To enforce a consistent polarity convention, we inverted waveforms whose maximum absolute deflection was positive, ensuring that all had a dominant negative initial peak, typical of somatic action potentials.

In Figure 5A, we illustrate the set of the eight morphological features extracted from each unit’s spike waveform defined as follows:

1. **Spike width** (trough-to-peak duration): reflects the time between the negative and positive peaks; typically shorter in interneurons.
2. **Half-width**: duration at half-maximal amplitude; another measure of spike duration and symmetry.
3. **Peak-to-trough amplitude ratio**: quantifies waveform asymmetry.
4. **Trough-to-peak amplitude ratio**: inverse of the previous, providing complementary asymmetry information.
5. **Repolarization slope**: steepest post-trough decline; faster in inhibitory cells.
6. **Area after minimum**: integral of the waveform post-trough, indicating repolarization extent (Trainito et al., 2019)
7. **1 – left peak**: asymmetry index based on pre-trough peak (Oghazian et al., 2021).
8. **Peak-to-trough ratio**: additional symmetry metric.

All features were z-scored across units to normalize variance and avoid scale-driven bias.

4.2.2 Clustering Methodology

To classify well-isolated single units into putative excitatory or inhibitory categories, we implemented two complementary unsupervised clustering approaches based on spike waveform features: one data-driven and one physiologically motivated.

Approach 1 – PCA-Based K-Means Clustering

In this approach, the eight z-scored waveform features were submitted to Principal Component Analysis (PCA) to reduce dimensionality while preserving maximal variance. The resulting low-dimensional representation was then clustered using the K-Means algorithm with $k = 2$ to separate units into two groups based on waveform morphology.

To better understand the axes of separation in PCA space, we examined the loading scores of each original feature onto the retained components. The features contributing most to the principal axes were then used to construct a reduced feature set. A secondary K-Means clustering ($k = 2$) was subsequently performed directly on this reduced set of three features, without dimensionality reduction, to assess their individual discriminative power.

Approach 2 – Physiologically-Guided K-Means Clustering

In parallel, we implemented a clustering strategy grounded in prior electrophysiological literature. Rather than using PCA, we directly applied K-Means clustering ($k = 2$) on three biologically validated features known to distinguish excitatory from inhibitory neurons (Barthó et al., 2004; Vigneswaran et al., 2011):

1. **Firing rate** — typically higher in interneurons.
2. **Peak-to-trough ratio** — generally lower in narrow-spiking inhibitory cells.
3. **Half-width duration** — shorter in fast-spiking neurons.

These features were z-scored and submitted to K-Means clustering in their native space. This strategy provides a biologically interpretable reference framework for putative cell-type classification.

4.2.3 Neuronal Population Dynamics

To quantify the global effect of 5-MeO-DMT on neuronal activity in the anterior insular cortex, we analyzed the firing rate of all units labeled as "good" by Kilosort, independently of their waveform characteristics. Spike times were binned in fixed-width intervals of 30 seconds, and spike counts per bin were computed for each unit across the full recording duration. Firing rate trajectories were aligned with respect to the injection event using normalized behavioral intervals defined for each mouse (before_injection and after_injection), as described earlier.

Z-score normalization and binning

For each unit, a firing rate z-score was computed per time bin, using the pre-injection period as baseline. Specifically, mean and standard deviation of the firing rate during the baseline window were used to normalize the binned spike counts over the entire session:

$$z_i(t) = \frac{N_i(t) - \mu_{baseline}}{\sigma_{baseline}}$$

With $N_i(t)$ its spike count, $\mu_{baseline}$ its mean firing rate before injection, and $\sigma_{baseline}$ the corresponding standard deviation.

This approach allows for inter-unit comparability, independent of absolute firing rates.

Thresholding and Classification of Unit Modulation

To classify functionally modulated neurons, we applied a statistical threshold based on a 99% confidence interval under Gaussian assumptions ($|Z| > 2.58$). This threshold was justified by the narrow distribution of Z-scores in saline animals, where nearly all units remained within ± 1 . Accordingly, units were categorized as:

- **Excited:** mean Z-score $> +2.58$
- **Inhibited:** mean Z-score < -2.58
- **No change:** Z-score within ± 2.58

Quantification of Post-Injection Modulation via Z-Scored Firing Rates

To quantify this modulation, we analyzed the Z-score distributions for all units (excluding one animal, F3880, due to insufficient data: only 6 units compared to >30 in all others), as well as for subgroups classified by modulation direction (**Figure 5H**). Units were categorized based on their

average Z-score in the post-injection window, using a threshold of $|Z| > 2.58$ corresponding to $p < 0.01$ under a standard Gaussian distribution.

5. Statistical analysis

5.1 Behavioral Data

Statistical comparisons of behavioral parameters—including total displacement, forward and backward movement, and movement ratios—were performed using **linear mixed-effects models (LMMs)**. Each model included `group` (5-MeO-DMT vs. saline), `interval` (10-minute time bins), and their interaction as **fixed effects**, with a **random intercept** for each mouse to account for inter-individual variability.

Models were fitted using the **Restricted Maximum Likelihood (REML)** method. The significance of fixed effects was assessed using **Wald Z-tests**, and post hoc comparisons were based on **estimated marginal means**, corrected for multiple comparisons using the **False Discovery Rate (FDR)**.

When variables violated normality assumptions, log-transformations were applied prior to modeling. Residuals were visually inspected to assess model validity.

5.2 Oscillatory Power

For each frequency band (delta, theta, alpha, beta, gamma), baseline-normalized power values were averaged per interval and per mouse. These values were analyzed using **separate LMMs for each band**, with `group`, `interval`, and their interaction as fixed effects, and a random intercept for each mouse.

Significant effects were evaluated using **Wald Z-tests**, and post hoc comparisons of marginal means were **FDR-corrected**. Model outputs included confidence intervals and estimated effect sizes (e.g., dB difference between groups).

FOOOF-derived spectral parameters (slope, offset, peak frequency, and peak power) were analyzed as change scores (post – pre). Between-group comparisons were performed using **Mann–Whitney U-tests**. Within-group changes were assessed via **paired permutation tests**. For peak features, additional LMMs were applied with `group` and `condition` (pre vs. post) as fixed effects and a random intercept for each mouse.

5.3 Single-Unit Activity

Firing rate Z-scores were computed per unit using baseline mean and standard deviation, then binned and averaged by interval. These values were analyzed using **LMMs** with `group`, `interval`, and `group × interval` as fixed effects and a random intercept per mouse.

Units were categorized as **excited**, **inhibited**, or **unchanged** based on post-injection Z-scores exceeding statistical thresholds ($|Z| > 2.58$). The distribution of modulation types between groups was compared using **Mann–Whitney U-tests**, with **FDR correction**.

No additional statistical modeling (e.g., correlation or causality analysis) was reported for single-unit dynamics in the current manuscript.

RESULTS

1. Behavioral analysis

1.1. Head-twitch response

In the 5-MeO-DMT group, head-twitch responses (HTRs) increased sharply immediately after injection, peaking within the first five minutes and then gradually declining over time (Figure 2A). In contrast, saline-treated mice displayed only rare and sporadic HTRs, without a consistent temporal pattern. This time course aligns with the expected pharmacokinetics of intraperitoneal 5-MeO-DMT and confirms its rapid central action and engagement of serotonergic pathways.

2.1. Locomotion

2.1.1 Displacement analysis

Representative locomotor trajectories are shown in Figure 2B for one mouse per group of the first 30min after injection. The saline mouse displays smooth, continuous paths distributed across the arena, reflecting organized exploratory behavior. In contrast, the 5-MeO-DMT mouse exhibits fragmented, looping trajectories with dense recurrences in central zones, interspersed with prolonged immobility. The color-coded overlay (green: forward; red: backward; gray: immobile; blue: turning) reveals a **notable prevalence of backward movement and immobility** during the first phase post-injection in the 5-MeO animal, rarely observed in controls.

To quantify these changes at a population level, we modeled total displacement, forward displacement, and the backward/(forward + backward) ratio using linear mixed-effects models (LMMs), with mouse identity as a random effect.

Analysis of **total displacement** (top panel, Fig. 2C) revealed a significant main effect of group (Coef. = 4060.37, SE = 1591.61, $z = 2.55$, $p = 0.011$), indicating that saline-treated mice traveled

significantly further than 5-MeO-DMT mice during the baseline interval (0–10 min). Within the 5-MeO group, displacement increased significantly in the 30–40 min (Coef. = 2770.33, $p = 0.002$) and 40–50 min intervals (Coef. = 2346.15, $p = 0.009$), consistent with a partial recovery of movement. However, negative group $\times$ interval interactions were also significant (e.g., -4744.50 for 30–40 min, $p < 0.001$), confirming that saline mice maintained high levels of movement throughout, while 5-MeO mice began from a suppressed state and gradually normalized.

Forward displacement followed a similar trend (middle panel, Fig. 2C). Saline mice exhibited greater forward movement at baseline (Coef. = 4791.75, SE = 1376.55, $z = 3.48$, $p < 0.001$), and 5-MeO animals displayed progressive increases at 30–40 min (Coef. = 3945.86, $p < 0.001$), 40–50 min (Coef. = 3505.64, $p < 0.001$), and 50–60 min (Coef. = 2844.66, $p = 0.001$). However, strong negative interactions (e.g., -6599.50 at 50–60 min, $p < 0.001$) highlight that saline mice remained consistently more active. A significant interaction at 20–30 min (Coef. = -2883.75 , $p = 0.025$) suggests that the **peak of locomotor inhibition** occurred within the first half hour after injection.

The **backward ratio** (middle panel, Fig. 2C) showed a prominent, transient increase in 5-MeO mice. The LMM revealed a strong main effect of group (Coef. = -0.221 , SE = 0.037, $z = -6.03$, $p < 0.001$), indicating that 5-MeO-DMT induces significantly more backward movement than saline. Negative time effects (all $p < 0.001$) and significant group $\times$ interval interactions (e.g., Coef. = $+0.267$ at 20–30 min) indicate that this behavior peaked early and declined thereafter.

2.1.2 Time-based movement classification

To complement displacement metrics, we classified each frame as **immobile**, **forward**, or **backward**, independent of movement amplitude, and computed the ratio of time spent in each state.

The **backward/moving** time ratio (top panel, Fig. 2D) showed a strong negative main effect of the saline group (Coef. = -0.392 , SE = 0.062, $z = -6.29$, $p < 0.001$), confirming that **5-MeO-DMT uniquely elicits backward behavior**, especially in the first 20 minutes. Group $\times$ interval interactions became significant from the 10–20 min bin onward ($p < 0.001$), and by 50–60 min, the backward ratio in 5-MeO mice had normalized to saline levels.

Similarly, the **immobile/total time** ratio (bottom panel, Fig. 2D) was significantly higher in 5-MeO mice at baseline (Coef. = -0.360 , $p < 0.001$). A significant time effect emerged at 30–40 min (Coef. = -0.162 , $p = 0.026$), with maximal mobility restored by 50–60 min (Coef. = -0.431 , $p < 0.001$). Significant group $\times$ interval interactions in these late bins indicate that **5-MeO-treated mice gradually regained normal movement patterns**, in contrast to the stable behavior of saline controls.

The forward movement ratio (Figure 2E) showed a robust main effect of group, with saline-treated mice displaying significantly higher levels of forward locomotion compared to 5-MeO-DMT mice across the session (Coef. = 0.366, SE = 0.073, $z = 5.00$, $p < 0.001$). This indicates a strong disruption of goal-directed movement following 5-MeO-DMT injection.

In 5-MeO-treated animals, forward movement increased progressively over time, reflecting a partial recovery of motor drive. Significant within-group improvements were detected in the 10–20 min ($p < 0.001$) and 20–30 min ($p < 0.001$) intervals compared to baseline. However, group $\times$ interval interactions were also significant in these time windows ($p < 0.001$), confirming that saline mice maintained high levels of forward motion while 5-MeO mice exhibited a delayed but significant rebound.

These findings reveal that 5-MeO-DMT induces a transient but structured disruption of spontaneous locomotion, marked by an initial suppression of forward movement, increased immobility, and the emergence of atypical behaviors such as backward walking. Although mobility gradually recovers over time, it remains below control levels during the early post-injection phase. This motor phenotype reflects not just a reduction in activity, but a reorganization of movement dynamics, suggesting altered sensory integration or disrupted internal body representation under the psychedelic state.

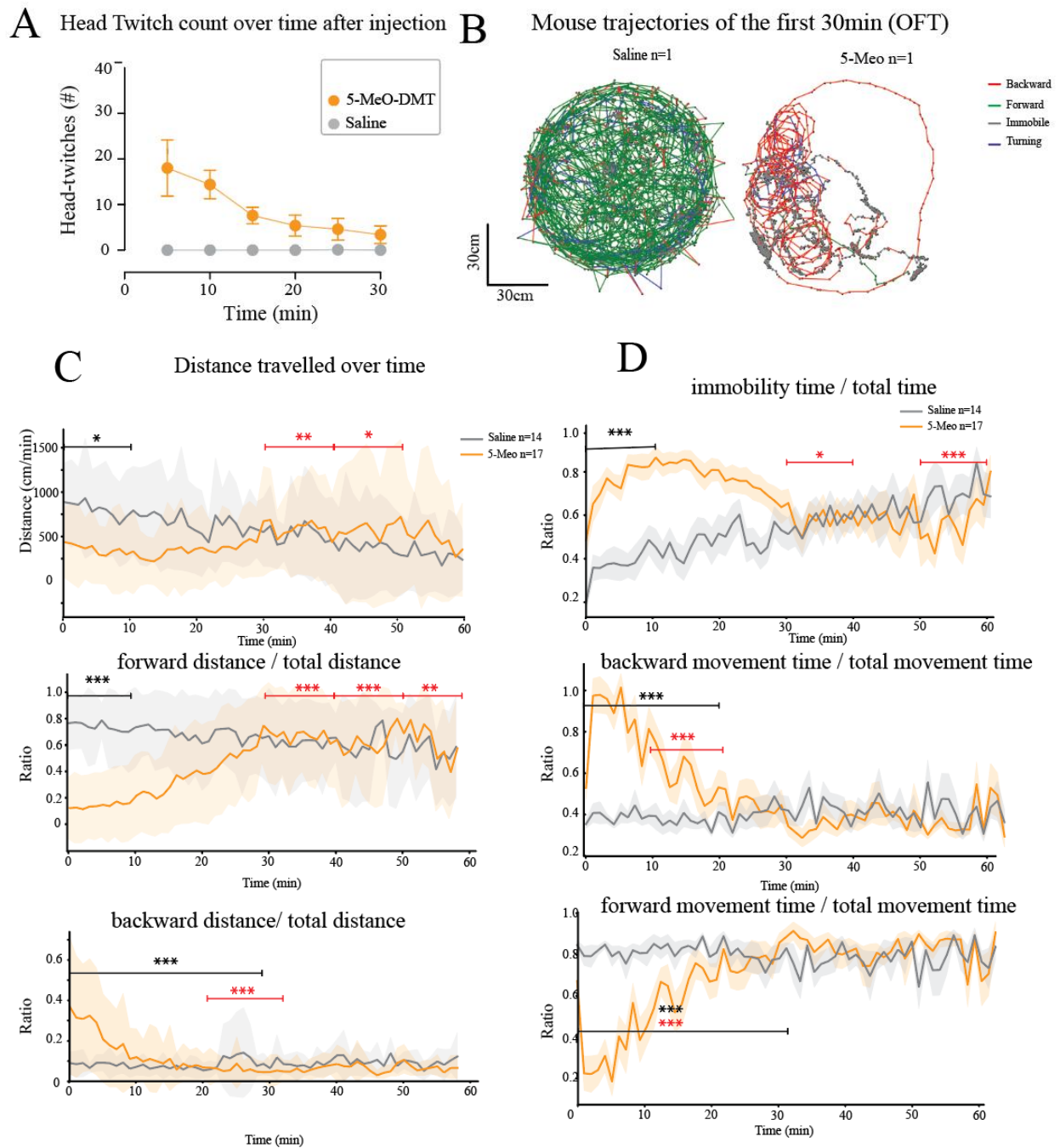


Figure 2. 5-MeO-DMT disrupts spontaneous locomotor activity and alters directional movement dynamics.

A. Mean number of head-twitches per 5-minute bin during the first 30 minutes post-injection in saline- and 5-MeO-DMT-treated mice. HTRs were detected using an electromagnetic coil system: a small magnet affixed to the mouse's ear induced measurable pulses during rapid head rotations, which were recorded and amplified via a copper-wire coil surrounding the behavioral arena. Automatic detections were subsequently validated by visual inspection using synchronized video recordings.

5-MeO-DMT induced a robust increase in head-twitches compared to saline, particularly during the first 10 minutes post-injection, with a gradual decay over time. No head-twitches were detected in saline-treated mice. Data are shown as mean $\pm$

SEM. B. Representative locomotor trajectories during the first 30 minutes of the open field test (OFT). Color-coded trajectories of a single saline-treated mouse (left) and a single 5-MeO-DMT-treated mouse (right), reconstructed from centroid tracking and movement classification. Forward movement is shown in green, backward in red, immobility in gray, and turning events in blue. The saline animal displays a dense and uniform coverage of the open field, dominated by forward motion. In contrast, the 5-MeO-DMT mouse exhibits spatially restricted, looping trajectories largely composed of backward and immobile segments, with minimal outward exploration. This qualitative contrast illustrates the profound alteration of navigation strategies and spatial engagement induced by 5-MeO-DMT, even in the absence of gross motor impairment.

C Linear displacement metrics extracted from centroid tracking over the 60-minute post-injection period. **Top panel:** Total distance travelled (cm/min). Saline-treated mice maintained stable displacement over time, whereas 5-MeO-DMT mice displayed a marked reduction during the initial 10-minute interval ($p = 0.011$), followed by a progressive increase at 30–40 and 40–50 minutes ($p = 0.002$, $p = 0.009$). However, significant group $\times$ time interactions indicate that 5-MeO animals remained overall hypoactive compared to controls. **Middle panel:** Ratio of forward distance to total distance. Saline animals exhibited consistently higher forward displacement, particularly in early intervals ($p < 0.001$). In 5-MeO-treated mice, forward ratio increased significantly across the session, with main effects of time in the 30–40, 40–50, and 50–60 minute bins ($p < 0.001$, $p < 0.001$, $p = 0.001$), indicating delayed but robust motor recovery. **Bottom panel:** Ratio of backward to total distance. 5-MeO mice showed a strong early increase in backward movement ($p < 0.001$), peaking around 20–30 minutes and declining thereafter. Saline mice exhibited minimal backward movement.

D Time-based classification of behavioral states: **Top panel:** Immobility ratio (time spent immobile / total time). 5-MeO-DMT significantly increased immobility during the first 10 minutes ($p < 0.001$), with reductions at 30–40 ($p = 0.026$) and 50–60 minutes ($p < 0.001$), marking a progressive reactivation of movement. **Middle panel:** Ratio of backward movement time to total movement time. A significant group effect confirmed that 5-MeO-DMT elicited enhanced backward responses early in the session ($p < 0.001$), with within-group decreases observed in the 10–20 minute bin ($p < 0.001$). By the final time bin (50–60 min), the backward profile had normalized to saline levels. **Bottom panel:** Ratio of forward movement time to total movement time. A strong main group effect showed that saline mice consistently engaged in forward movement more than 5-MeO mice across the session ($p < 0.001$). Intra-group recovery in the 5-MeO group was significant in the 10–20 and 20–30 min intervals ($p < 0.001$), while group $\times$ time interactions confirmed a divergent trajectory compared to saline animals.

3. Spectral Analysis

3.1 Power Spectral Density (PSD)

3.1.1 Global Normalized Power Spectrum

We first examined the **average post-injection power spectrum** across frequencies for each group (Fig. 3A). This spectrum represents the **global modulation** of frequency content during the full post-injection period compared to baseline.

In the **5-MeO-DMT group**, a **clear increase in spectral power** is observed between **40 and 80 Hz**, corresponding to the **gamma band**. This enhancement appears selective and pronounced, consistent with prior reports of psychedelic-induced gamma oscillations.

Additionally, **subtle increases** are present in **lower frequency ranges** (1–8 Hz), suggesting that delta and theta power may also be modulated, although to a lesser extent. In contrast, **alpha (8–11 Hz) and beta (11–30 Hz) bands** show little change.

An interesting **narrowband artifact** appears just **before injection** in several recordings at 50 Hz. This is likely attributable to **mouse handling noise** or **defective grounding** at the time of injection, not of physiological origin. This motivated the decision to focus on **time-resolved, band-specific analyses** rather than relying solely on global spectral averages.

3.1.2 Temporal Evolution of Canonical Bands

Key trends were observed In gamma, the 5-MeO-DMT group exhibits a rapid and pronounced rise starting immediately after injection, followed by a slow decay. In delta, a broader and more sustained increase is present, peaking around 10–20 minutes post-injection. Alpha power shows a modest but consistent decrease from 10 minutes onward and finally Theta and beta bands appear largely unmodulated in this resolution.

These dynamics reinforce the hypothesis of selective frequency modulation, particularly in gamma and delta bands, and highlight the transient nature of the drug's effect on high-frequency activity.

3.1.3 Group-Level Median Spectrograms

To visualize the **full spectral and temporal profile** of each condition, we looked at the **median spectrogram** per group shown in **Figure 3D**.

In the **5-MeO-DMT panel**, we observe first a **strong, temporally precise increase** in gamma band power following injection and also a **dense accumulation of power in the delta band**, lasting longer and with broader temporal structure with **minimal structure in the alpha band**, supporting the subtlety observed in the curves.

In contrast, the **Saline condition** exhibits **relatively stable power** across all bands, with no systematic increase in any frequency domain.

This visualization confirms that the major modulations observed in the 5-MeO condition are real, time-locked, and frequency-selective, particularly in gamma and delta bands.

While the spectrograms and time-series plots reveal **striking qualitative differences** in oscillatory dynamics between groups, a formal quantification is needed to assess **When** these modulations occur, **Which** frequency bands are affected, And whether these effects are **specific to 5-MeO-DMT** or also present in the control group.

To achieve this, we computed the **average power per band and time interval**, and analyze temporal and group-level effects.

3.1.4 Frequency-Specific Effects of 5-MeO-DMT

The **gamma band (30–80 Hz)** exhibited the most prominent and selective modulation. In 5-MeO-DMT mice, power rose sharply during the first 30 minutes post-injection, with increases of +2.50 dB (0–10 min), +2.23 dB (10–20 min), and +1.50 dB (20–30 min), all highly significant ($p < 0.001$). No comparable change was observed in saline controls. A significant group $\times$ interval

interaction (e.g., -2.41 ± 0.60 at 0–10 min) confirmed that this gamma enhancement was specific to the drug condition and time-locked to its acute effects.

In the **delta band (1–3 Hz)**, spectral power also increased markedly in the 5-MeO group, reaching approximately +3.4 dB throughout the 0–30 min post-injection window ($p < 0.001$). These effects were again absent in the control group and accompanied by strong interaction terms, pointing toward a sustained low-frequency shift. However, unlike gamma, this enhancement appeared less precisely time-locked, and may reflect **a broader alteration of arousal or global cortical state**.

The **alpha band (8–11 Hz)** showed a more modest but consistent decrease in power, emerging around 10 minutes post-injection and persisting through 30 minutes (-1.2 dB on average, $p < 0.01$). While this suggests a potential disinhibition of sensory or thalamo-cortical rhythms, the lack of a significant interaction with group implies that this suppression was not reliably drug-specific and may be more variable across subjects.

Changes in the **theta band (3–8 Hz)** were subtler: 5-MeO-DMT induced slight power increases in the first 20 minutes (+0.88 and +0.72 dB), with marginal statistical significance ($p < 0.05$). However, these effects did not significantly differ from saline, indicating a more diffuse or secondary modulation, possibly reflecting background network reorganization rather than a primary oscillatory signature.

Finally, **beta-band activity (11–30 Hz)** remained stable across all timepoints and conditions ($p > 0.2$), with no significant main or interaction effects detected. This suggests that beta oscillations, typically associated with motor processes and sensorimotor gating, were not engaged or modulated under the present conditions.

In summary, these results show that:

Gamma and delta bands are **most robustly and specifically modulated** by 5-MeO-DMT, whereas **Alpha and theta** exhibit **weaker or nonspecific changes**. Only **Beta** remains **stable**.

Conclusion – Are These Oscillations Structured or Nonspecific?

The analyses described above demonstrate a **global increase in oscillatory power**, particularly in gamma and delta bands. However, this raises a deeper mechanistic question:

Are these oscillations meaningful? Or do they merely reflect a non-specific increase in LFP signal amplitude?

If the **total energy of the LFP increases** due to enhanced synaptic drive or arousal, then all frequency bands may rise together — a phenomenon known as **broadband shift**. In this case, gamma and delta rises might **not reflect true oscillations**, but simply a global elevation of background activity.

Alternatively, if **specific peaks emerge** in frequency bands (e.g., gamma at ~60 Hz), this would suggest **functional synchronization**, e.g., reflecting altered perception, network binding, or cortical resonance.

To disentangle these scenarios, we next decomposed the power spectra using **spectral parameterization (FOOOF)**, separating **Aperiodic components** (1/f slope and offset), from **true oscillatory peaks** (center frequency, amplitude, bandwidth).

This method allows us to test whether gamma increases are **structurally specific** — and not just broadband artifacts.

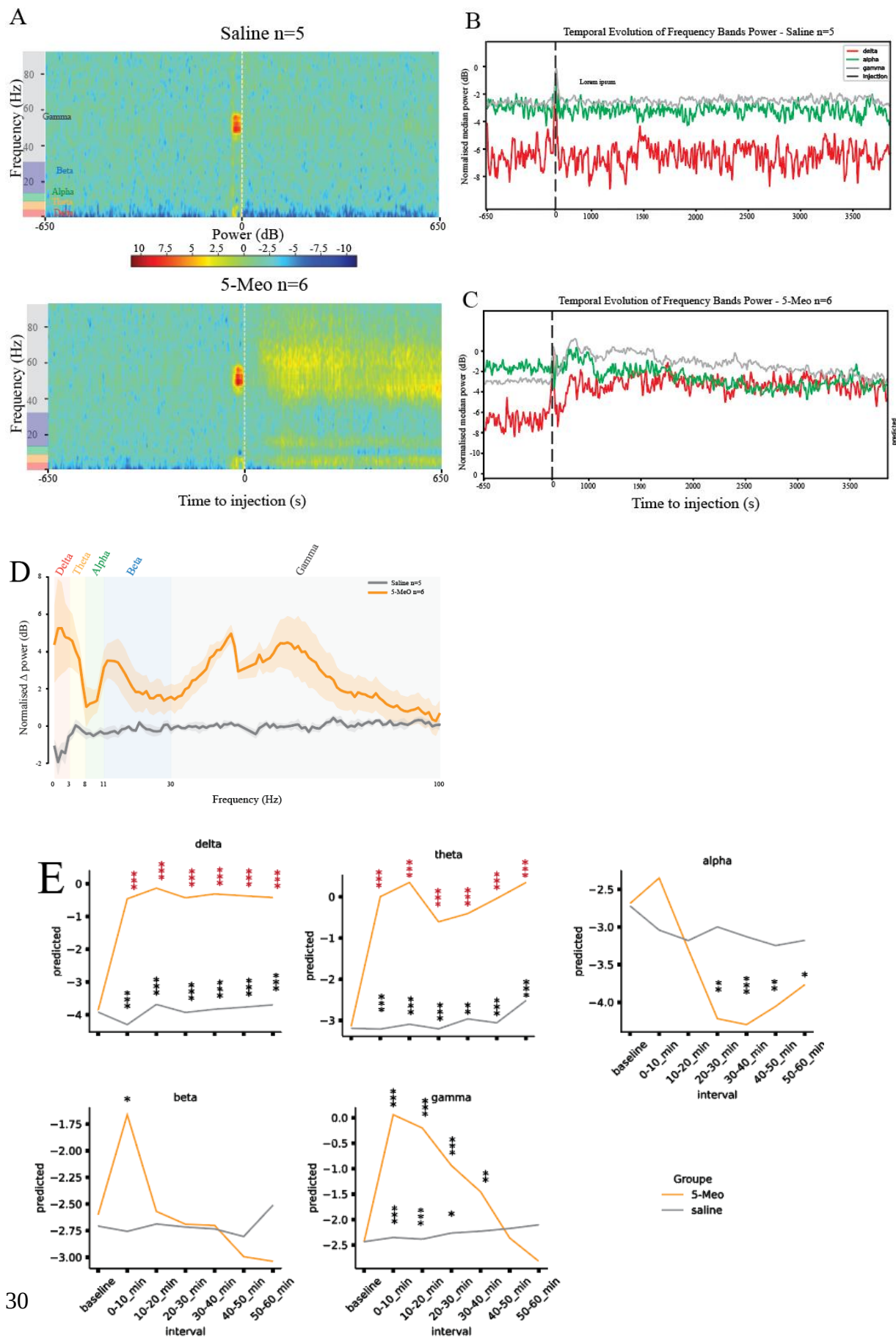


Figure 3 Power Spectral Density Modulated by 5-Meo-DMT in time. A. Group-level median spectrograms of baseline-normalized LFP power aligned to the moment of injection. Time–frequency representations (1–90 Hz) display local field potential (LFP) power in decibel units relative to a pre-injection baseline. Spectrograms are aligned to the time of intraperitoneal injection ($t = 0$ s, dashed line) and represent the group-level median across animals (Saline: $n = 5$; 5-MeO-DMT: $n = 6$). Each individual spectrogram was first normalized to the animal’s baseline, then averaged within group. In the saline condition (top), LFP power remains globally stable across time and frequency. In the 5-MeO-DMT condition (bottom), distinct modulations of spectral power are visible. The color scale indicates power change in decibels (dB) relative to baseline. Frequency bands are indicated on the left for reference. **e B–C. Time-resolved evolution of baseline-normalized frequency band power following injection.** Median trajectories of local field potential (LFP) power (in dB relative to pre-injection baseline) are shown for delta (1–4 Hz, red), alpha (8–12 Hz, green), and gamma (30–80 Hz, gray) frequency bands. Power values were extracted from spectrograms of each animal, smoothed with a 30-second moving average, and averaged within group. Panel B corresponds to saline-treated animals ($n = 5$), and panel C to 5-MeO-DMT-treated animals ($n = 6$). Time is aligned to the injection ($t = 0$ s, dashed vertical line). Power is expressed as baseline-normalized change (dB), computed per mouse prior to averaging. No shading is shown. **D. Baseline-normalized power spectra averaged over the post-injection period for each group.** Power spectral density (PSD) curves were computed from the 0–600 s post-injection interval, normalized to each animal’s baseline (in dB), and averaged across animals (Saline: $n = 5$; 5-MeO-DMT: $n = 6$). The x-axis displays frequency (1–90 Hz), and the y-axis indicates power change in decibels. Shaded regions highlight canonical frequency bands: delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–80 Hz). Each PSD reflects the mean of individual mouse spectra, computed over the entire post-injection window. No smoothing beyond spectral estimation was applied. **E. Linear mixed-effects modeling of spectral power across time and frequency bands.** Baseline-normalized power (in dB) is shown for canonical frequency bands—delta (1–3 Hz), theta (3–8 Hz), alpha (8–11 Hz), beta (11–30 Hz), and gamma (30–80 Hz)—averaged within successive 10-minute intervals post-injection. Curves represent model-predicted group means from linear mixed-effects models (LMM), with 5-MeO-DMT (orange) and saline (gray) groups plotted separately (Saline: $n = 5$; 5-MeO-DMT: $n = 6$). Power was normalized per animal to pre-injection baseline prior to modeling. Time axis is segmented into 10-minute bins (0–60 min). Black asterisks mark significant inter-group differences (5-MeO-DMT vs. saline); red asterisks denote significant within-group deviations from baseline in the 5-MeO-DMT group. Statistical significance was assessed via LMM with FDR correction.

3.2 Fitting Oscillation and One Over F (FOOOF)

3.2.1. Visual Decomposition Example (Figure 4A)

Panels **A (top)** and **A (bottom)** of Figure 4 illustrate the FOOOF decomposition for one representative 5-MeO-DMT mouse:

Before injection (top) the spectrum exhibits a **smooth $1/f$ decay** with a visible alpha peak around ~8–10 Hz. The aperiodic background (dashed blue) and full fit (orange) follow the observed spectrum closely. **After injection (bottom)** the spectrum is **elevated across frequencies**, particularly above 30 Hz. The alpha peak disappears, and **distinct peaks in the gamma band** emerge (~40–70 Hz), accompanied by a **steeper slope and higher offset**.

3.2.2. Aperiodic Dynamics – Slope and Offset (Figure 4B–D)

Slope

The analysis of the *slope* parameter (Figure 4B), which represents the steepness of the aperiodic spectral decay in log–log coordinates, reveals divergent dynamics between the two experimental groups. In the 5-MeO-DMT group, the slope becomes significantly steeper after injection (mean

before injection: 1.251; after injection: 1.461), representing an average increase of 0.210. In contrast, a slight decrease is observed in the Saline group (from 1.337 to 1.263, $\Delta = -0.074$). This directional difference suggests an enhancement of the 1/f profile exclusively under 5-MeO-DMT. Although statistical tests on the slope delta do not exceed conventional significance thresholds (Mann-Whitney U $p = 0.1775$; permutation test $p = 0.1135$), the observed effect size is notable (Cohen's $d = 0.97$), indicating a robust trend likely attenuated by inter-subject variance.

Offset

The *offset* parameter (**Figure 4C**), corresponding to the baseline amplitude of the aperiodic spectrum, increases markedly in the 5-MeO-DMT group after injection (from 2.687 to 3.243, $\Delta = +0.555$), whereas a modest decrease is observed in the Saline group (from 2.850 to 2.719, $\Delta = -0.132$). This divergence is statistically significant across all tests applied (Mann-Whitney U $p = 0.0043$), with a very large effect size (Cohen's $d = 1.44$). The increase in offset suggests a generalized enhancement of background spectral activity under 5-MeO-DMT, consistent with increased excitation or widespread disinhibition of cortical circuits.

Slope–Offset Correlation

To better understand the relationship between treatment-induced changes in aperiodic spectral dynamics, a correlation analysis was performed on the *delta* values of the FOOOF parameters **Figure 4D**—that is, the difference after minus before injection for each mouse. The analysis reveals an extremely strong correlation between delta offset and delta slope: mice showing an increase in spectral slope systematically exhibit a parallel increase in offset level, and vice versa. This association is confirmed by a very high Pearson correlation ($r = 0.96$, $p = 3.625 \times 10^{-6}$).

Strikingly, the graphical representation of this relationship reveals a clear separation between experimental groups: all 5-MeO-DMT mice cluster on the right side of the scatter plot, corresponding to a dual increase in slope and offset, whereas all Saline mice cluster on the left, reflecting near-zero or negative deltas. This configuration suggests that the co-modulation of slope and background amplitude constitutes a robust spectral signature of 5-MeO-DMT's pharmacological effect, with a coherent and distinguishable dynamic at the individual level. This suggests a **shared underlying mechanism**, possibly involving global shifts in cortical excitability or gain control.

2.2.3. Oscillatory Peaks – Frequency and Power (Figures 4E–G)

Beyond the aperiodic background, FOOOF identifies **periodic peaks**—true rhythmic oscillations that rise above the 1/f structure. We analyzed both the **center frequency** of the most prominent peak (0–80 Hz range), and its **relative power** (height above the background), within alpha and gamma bands.

Peak Frequency.

We modeled peak frequency using a **linear mixed-effects model (LMM)** with fixed effects for condition (before/after), group (5-MeO vs. Saline), and their interaction.

The model revealed a significant main effect of condition ($p < 0.0001$), indicating that peak frequencies increased after injection, as well as a group effect ($p < 0.0001$), showing that 5-MeO mice had overall higher-frequency peaks. Crucially, a significant interaction ($p = 0.001$) confirmed that this upward shift was specific to the psychedelic condition. This is illustrated in Figure 4E, where dominant peaks in 5-MeO animals migrate upward post-injection, clustering around the gamma range (~60 Hz), while saline-treated animals exhibit no such displacement.

Gamma Peak Power (30–80 Hz)

We then examined peak power in the gamma band (30–80 Hz), again using a linear mixed-effects model. The analysis revealed a strong main effect of condition ($p < 0.0001$) and a significant condition $\times$ group interaction ($p < 0.0001$), indicating that gamma peak amplitude increased significantly after 5-MeO-DMT injection, but remained unchanged in saline controls. As shown in Figure 4F, gamma peak power rises substantially in 5-MeO mice post-injection, whereas it stays flat in the control group. These findings confirm that gamma activity under 5-MeO-DMT is not purely a result of broadband elevation (as observed in the PSD), but also includes the emergence of true, structured high-frequency oscillations—likely reflecting enhanced local synchronization or altered perceptual integration

Alpha Peak Power (7–11 Hz)

In contrast, alpha peak power (7–11 Hz) displayed a selective suppression in 5-MeO mice. The linear model identified significant main effects of condition and group (both $p < 0.0001$), and a significant interaction ($p < 0.0001$), indicating that alpha suppression was specific to the psychedelic group. As visualized in Figure 4G, alpha peaks observed before injection virtually disappear post-injection in 5-MeO animals, while they persist in saline controls. This suppression is consistent with previous findings in both animal and human studies and may reflect a disruption of thalamo-cortical rhythms or the breakdown of resting-state network coherence, both commonly associated with psychedelic-induced changes in sensory gating and cortical desynchronization

3.2.4. FOOOF conclusion

The FOOOF decomposition provides mechanistic insight into the spectral changes induced by 5-MeO-DMT, revealing a reorganization that involves both the aperiodic background and rhythmic peaks. On the aperiodic side, treated mice show a significant increase in offset, indicating a global enhancement of spectral power, and a non-significant but consistent increase in slope, absent in controls. The strong correlation between these two parameters suggests a coordinated shift in cortical tone, potentially reflecting changes in excitation/inhibition balance or neuromodulatory drive.

On the rhythmic side, gamma-band peaks emerge or intensify specifically after 5-MeO-DMT (Fig. 4F), confirming that gamma enhancement observed in PSDs is not merely broadband but reflects structured synchrony. In contrast, alpha peaks collapse in treated animals post-injection while persisting in controls (Fig. 4G), suggesting a disruption of thalamo-cortical idling or sensory gating. Finally, the dominant peak frequency shifts upward only in the 5-MeO group (Fig. 4E), reflecting a reorganization of cortical rhythms toward higher-frequency dynamics—a likely electrophysiological signature of the transition from baseline to altered states of consciousness.

3.3 The Spectral Signature of 5-MeO-DMT: Integrating PSD and FOOOF

Taken together, the findings from PSD and FOOOF analyses converge on a **coherent and multidimensional spectral phenotype** induced by 5-MeO-DMT.

From the **PSD perspective**, we observe an increase in power across multiple frequency bands—most notably in **gamma** and **delta**—alongside suppression in **alpha**. Time-resolved analyses show that these changes are transient but tightly locked to the post-injection phase.

FOOOF complements and refines the interpretation drawn from PSD analyses by revealing the spectral structure underlying the observed power changes. It confirms that the gamma-band enhancement is not solely attributable to broadband amplification, but instead reflects the emergence of genuine oscillatory peaks. It also shows that the increase in low-frequency power co-occurs with a steeper spectral slope, indicating a structured redistribution of spectral content rather than diffuse noise. Finally, FOOOF quantifies a dual modulation of offset and slope—present only in 5-MeO-DMT-treated animals—which defines a distinctive shift in aperiodic background activity and represents a robust spectral hallmark of the psychedelic state. In essence, 5-MeO-DMT induces a **spectral fingerprint** characterized by:

1. **Background destabilization:** increased offset and slope (FOOOF),
2. **Gamma activation and alpha suppression:** rhythmic reorganization (PSD + FOOOF),
3. **Specificity** to pharmacological condition: absent in saline, reproducible across individuals.

This spectral reconfiguration reflects a shift in brain state that likely underlies the drug's effects on perception, self-processing, and consciousness. The joint use of PSD and FOOOF reveals not just that the brain is more active under 5-MeO-DMT—but that **its activity is fundamentally reshaped**.

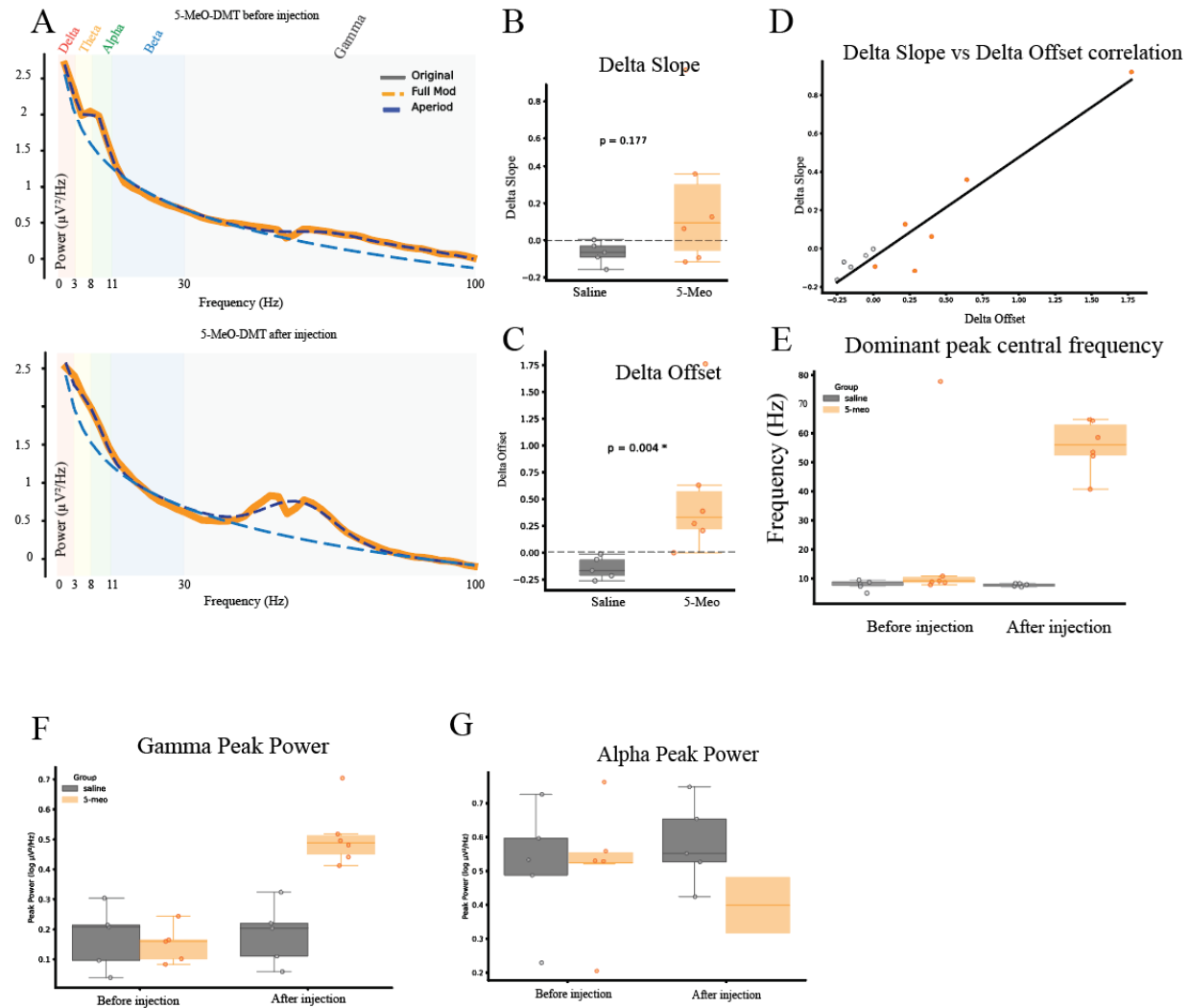


Figure 4 Figure 4. Spectral decomposition reveals dissociable effects of 5-MeO-DMT on aperiodic and oscillatory components. **A. Decomposition of power spectra into aperiodic and periodic components using FOOOF.** Representative log-log power spectral densities (1–100 Hz) are shown for one animal in the 5-MeO-DMT group, computed from the pre-injection (top) and post-injection (bottom) periods. Raw spectra (black traces) were modeled using the FOOOF algorithm, which separates the aperiodic component (blue dashed line) from narrowband periodic peaks (orange). The fits were performed in the 1–100 Hz range. Frequency bands are color-coded for reference. This decomposition enables characterization of changes in both the broadband background activity and superimposed rhythmic components. **B–C. Group-level comparison of aperiodic slope and offset following 5-MeO-DMT injection.** Panels show boxplots of aperiodic spectral parameters extracted from FOOOF fits (1–100 Hz), computed post-injection and grouped by treatment (Saline: $n = 5$, gray; 5-MeO-DMT: $n = 6$, orange). (B) Delta-band slope values (1–3 Hz) are plotted for each animal. The horizontal dashed line indicates zero slope (log-log linearity). No significant difference is observed between groups ($p = 0.277$). (C) Aperiodic offset (y-intercept of the log-log spectrum) is shown across the full spectral range. A significant increase in offset is observed in the 5-MeO-DMT group compared to saline ($p < 0.05$), indicating elevated broadband power. Each point corresponds to one animal; boxes show median and interquartile range. Statistical comparisons were performed using appropriate group-level tests; exact p -values are reported above the plots. **D. Correlation between aperiodic slope and offset following 5-MeO-DMT injection.** Scatterplot showing the relationship between the aperiodic slope and offset parameters extracted from FOOOF fits in the 5-MeO-DMT group ($n = 6$). Each point represents one animal. A linear regression line is shown in black. Both slope and offset were computed from post-injection spectra (1–100 Hz). The analysis illustrates the covariation of these two aperiodic features within subjects following drug administration. **E–G. Group-level comparison of peak frequency and power in dominant oscillatory components.** Peak parameters were extracted from FOOOF-identified periodic components in post- and pre-injection periods, for each animal.

and condition (Saline: $n = 5$, gray; 5-MeO-DMT: $n = 6$, orange).(E) Dominant peak center frequency is shown (in Hz), highlighting shifts in peak location across conditions.(F) Peak power of the dominant gamma component (30–80 Hz) is plotted for each group before and after injection. A marked increase is observed post-injection in the 5-MeO-DMT group. (G) Peak power in the alpha band (8–11 Hz) is shown across timepoints and groups. Each dot represents one animal; boxplots indicate median and interquartile range. Comparisons reflect within-group and between-group differences in periodic peak structure, estimated independently for each animal and timepoint.

4. Single-unit analysis

4.1 PCA-Based Clustering Results

4.1.2 Approach 1 – PCA + K-Means Clustering

Using the full set of waveform features, Principal Component Analysis revealed that the first three components accounted for 71% of the total variance (PC1: 30%, PC2: 23%, PC3: 18%) (Figure 5B). K-Means clustering ($k = 2$) applied in this reduced space identified two populations (Figure 5C), with a markedly unbalanced distribution:

- Cluster 0: 71 units
- Cluster 1: 510 units

To investigate the basis of this separation, feature loadings were examined, revealing that `area_after_min`, `repolarization_slope`, and `peak_to_trough_ratio` contributed most strongly to the principal axes (weighted by explained variance). Clustering performed directly on these three features (without PCA) produced a distinct partition:

- Cluster 0: 498 units
- Cluster 1: 83 units

4.1.3 Approach 2 – Physiologically-Guided K-Means Clustering

An alternative clustering based on three canonical features (`firing_rate`, `peak_to_trough_ratio`, `half_width_duration`) yielded two populations with a similarly skewed distribution (Figure 5D):

- Cluster 0: 508 units
- Cluster 1: 73 units

4.1.4 Cross-Validation and Agreement Between Methods

To evaluate the consistency of the classifications, we computed confusion matrices comparing labels from each approach. Because K-means labels are arbitrary (0/1), we applied label alignment to maximize agreement.

Comparison	Matching Units	% Agreement
PCA vs Physiological	495 / 581	85.2%
PCA vs Top Features	501 / 581	86.2%
Top Features vs Physio	563 / 581	96.9%
All 3 methods agreed	489 / 581	84.2%

To assess the consistency between classification methods, we first examined two illustrative examples. Figure 5E shows the confusion matrix comparing PCA-based clustering with physiologically defined categories across all recorded units. The majority of units were consistently classified—466 as excitatory and 29 as inhibitory—while only a minority showed mismatches (42 and 44 units), yielding an overall agreement of 85.2%.

At the individual level, Figure 5F details the agreement rates per mouse. Values ranged from 69.2% to 100%, with a mean of $85.6\% \pm 9.0\%$, demonstrating that this convergence is not limited to group-level effects but holds across subjects.

These results indicate that PCA-derived features such as repolarization slope and peak-to-trough ratio reliably reflect canonical physiological distinctions like spike width and firing rate. The strong overlap between the two approaches supports the robustness and generalizability of the classification framework.

4.1.5 Interpretation and Cluster Identity

The two clustering strategies revealed complementary yet converging patterns of unit classification. The PCA-based approach emphasized features related to post-spike repolarization—particularly `area_after_min`, `repolarization_slope`, and `peak_to_trough_ratio`—which likely capture subtle morphological variability, possibly influenced by recording geometry or noise structure. In contrast, the physiologically guided method relied on firing rate and spike width parameters, offering a more direct but narrower view of functional identity.

Despite these differences, all three clustering solutions (PCA space, top 3 features, and physiological features) converged on a shared core: 489 units (84%) were consistently assigned to the same cluster across methods (Figure 5E). This substantial agreement supports the presence of a robust and well-separated neuronal population. The remaining 92 units were assigned inconsistently, likely reflecting ambiguous waveforms, mixed cell identities, or marginal feature values.

Based on the directionality of separation and known electrophysiological markers, the larger cluster—comprising 510 to 538 units depending on the method—was interpreted as representing **putative excitatory neurons**, characterized by broader spike waveforms and lower firing rates. Conversely, the smaller cluster (43–71 units) exhibited narrower spikes, steeper repolarization slopes, shorter durations, and higher firing rates, consistent with **putative inhibitory interneurons** (Figure 5C–D).

These proportions are biologically plausible. The PCA-based classification identified approximately 12% inhibitory units (71/581), while the physiological clustering yielded 7.4% (43/581). Both values fall within the expected cortical range of 15–25% inhibitory neurons reported in the literature (e.g., *Folia Biologica*, 2021, Vol. 67, No. 6), noting that extracellular waveform clustering may underestimate fast-spiking interneurons due to their lower amplitude and ambiguous morphology.

These interpretations are further supported by the 3D visualization of the most discriminative features, which revealed a clear geometric separation between a compact, high-ratio excitatory cluster and a more dispersed, steep-slope inhibitory population.

While both classification schemes offer biologically meaningful results, we retained the PCA-based K-Means model for all subsequent analyses. In addition to its close alignment with canonical markers, this model captured a broader portion of waveform variability, integrating complex interactions between multiple features. This makes it more suitable for downstream comparisons involving graded or heterogeneous neuronal identities within the anterior insular cortex.

Effect of 5-MeO-DMT on Population Activity

When considering all units, 5-MeO-DMT induced a marked shift toward inhibition, with a mean Z-score of -2.49 compared to $+0.19$ in saline. Among positively modulated units ($Z > 0$), the 5-MeO group showed stronger activation (mean $Z = +3.77$) than saline (mean $Z = +0.72$), indicating that excitatory responses, while rare, were more pronounced. For negatively modulated units ($Z < 0$), inhibition under 5-MeO was considerably stronger (mean $Z = -3.66$) than in saline (mean $Z = -0.51$). All comparisons yielded highly significant p-values ($p < 0.001$), confirming a dual and robust modulation effect. These results indicate that 5-MeO-DMT profoundly suppresses population activity in the aIC, while selectively enhancing firing in a minority of neurons—suggesting a global inhibitory shift alongside targeted excitation.

The proportion of modulated units is summarized in **Figure 5I**. In the saline group, 99.2% of units remained unmodulated, with only 0.8% classified as excited and none as inhibited. By contrast, the 5-MeO-DMT group exhibited a striking shift in distribution: 5.8% of units were excited, 56.8%

inhibited, and 37.4% unchanged. This reconfiguration was statistically significant, as confirmed by Mann–Whitney U-tests (FDR-corrected $p = 0.0097$ for all class comparisons).

To assess whether modulation patterns were associated with neuronal subtype, we analyzed unit classes within each waveform-defined population. Among **putative GABAergic (inhibitory)** units ($n = 51$), 54.9% were inhibited and 3.9% excited. In **putative glutamatergic (excitatory)** units ($n = 291$), 57.7% were inhibited and 6.2% excited. This similarity in modulation proportions suggests that 5-MeO-DMT exerts its effects broadly across neuronal populations, with no clear preference for a given subtype.

Notably, both excited and inhibited units were observed in each cluster, indicating that waveform-based classification alone does not predict the direction of pharmacological modulation. This may reflect the limitations of morphological proxies or the diffuseness of 5-MeO-DMT's action across cortical circuits.

Temporal Dynamics of Neuronal Response to 5-MeO-DMT

Having established that a large proportion of anterior insula (aIC) units are inhibited following 5-MeO-DMT injection, we next turned to the **temporal evolution of the neuronal response** to better understand how population-level dynamics unfold over time.

Across the whole recorded population (Figure 5J), firing rates in the 5-MeO-DMT group show a marked and sustained suppression relative to baseline, while saline-treated mice display no such inhibition—instead exhibiting a transient increase in firing. This **global inhibition** likely reflects the high proportion of suppressed units observed in our classification. The pharmacological specificity of this suppression highlights the robust impact of 5-MeO-DMT on aIC network excitability.

However, this population-wide drop conceals more nuanced dynamics at the subpopulation level. When parsing the response by functional classification (Figure 5K), **two distinct temporal trajectories emerge**. Units classified as "inhibited" exhibit a sharp and stable reduction in firing that persists throughout the post-injection period—a pattern consistent with a **long-lasting shutdown**. In contrast, "excited" units show a rapid and strong **initial increase** in activity, which gradually wanes over time. This divergence reveals that **the global inhibitory profile is largely driven by the dominant pool of stably suppressed neurons**, but that **a subset of neurons undergoes a more complex, multiphasic modulation**.

This observation raises mechanistic questions: is the early excitation of these units a direct effect of 5-MeO-DMT, for example via 5-HT_{2A} receptor activation on excitatory cells? Or does it result from a **disinhibition mechanism**, where the silencing of local inhibitory neurons lifts suppression on downstream targets?

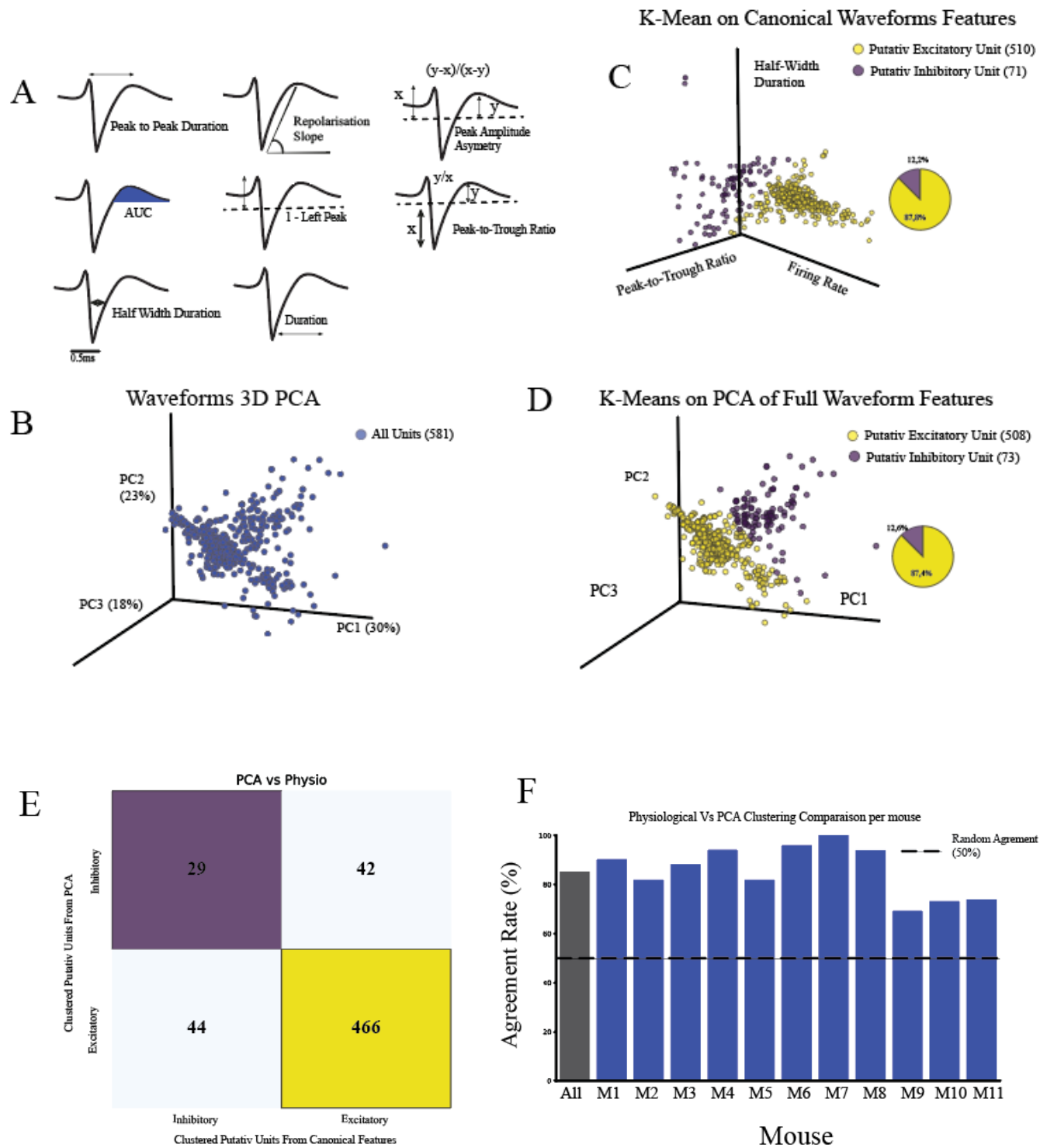
Altogether, this analysis highlights how 5-MeO-DMT not only dampens overall firing in the aIC, but reshapes the balance and timing of excitation and inhibition across the network. These findings reinforce the central role of E/I balance in shaping cortical dynamics under psychedelics, and

illustrate how network-level interactions modulate population responses in a non-uniform, time-dependent manner.

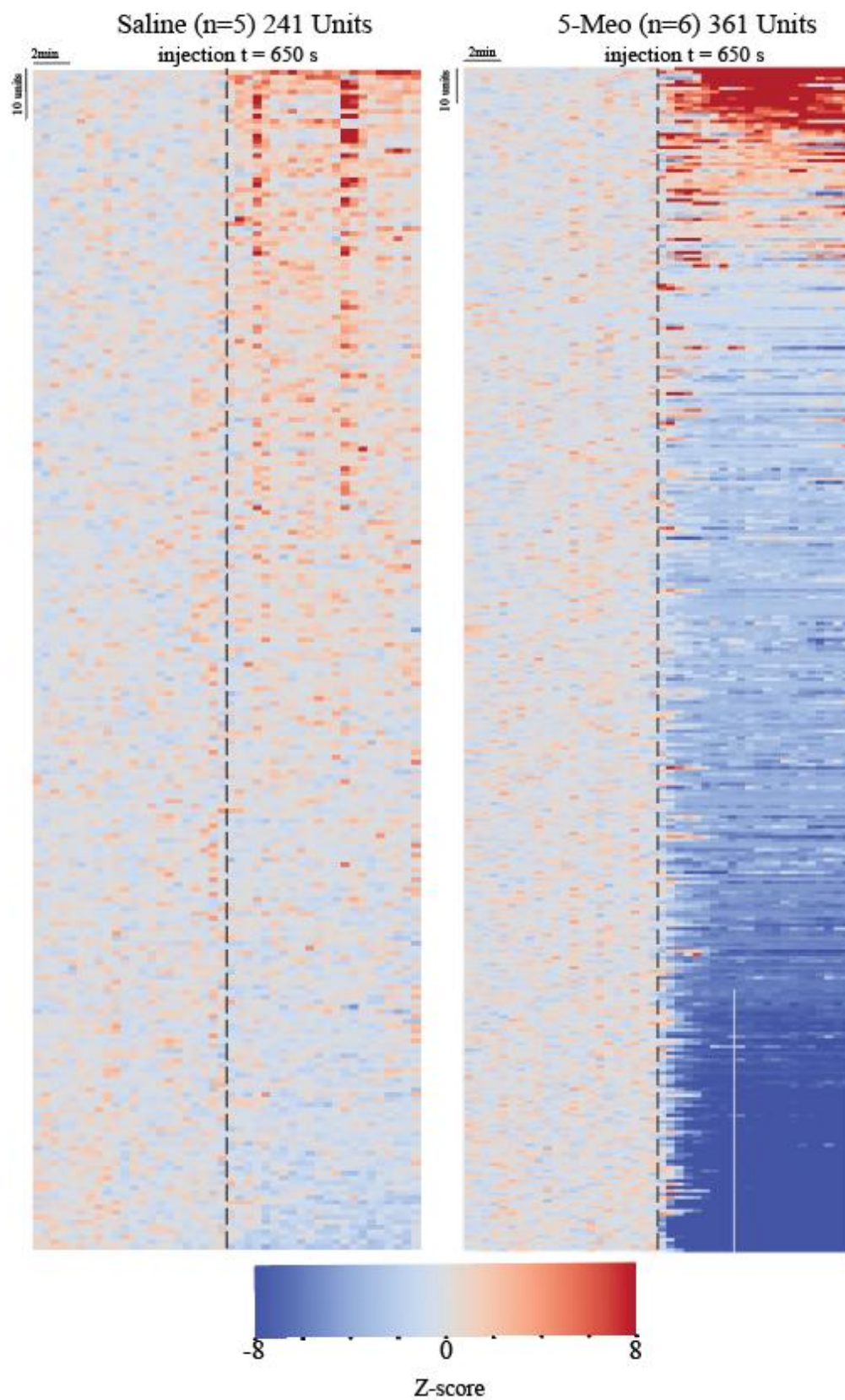
To further explore the **functional consequences of these firing patterns**, we next compared the evolution of neuronal activity with the dynamics of local field potential (LFP) oscillations. Specifically, we examined the **temporal profile of gamma-band power**—a frequency range often linked to cortical excitation and local processing—and asked how it related to the firing rate of excited units.

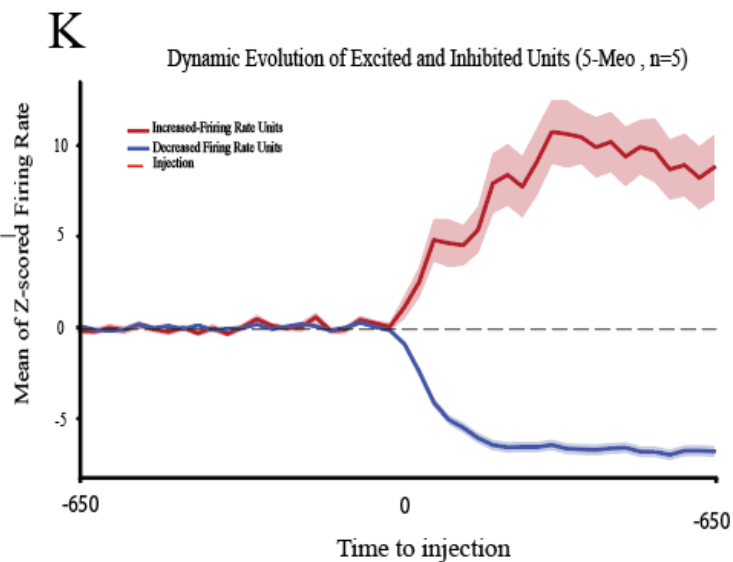
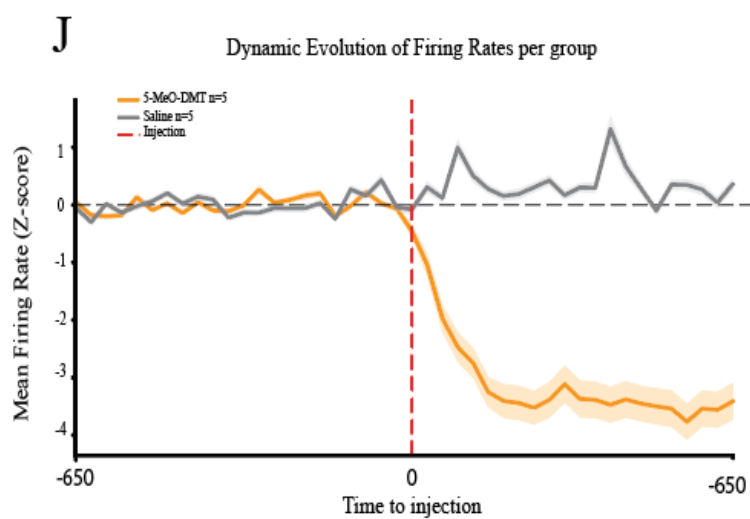
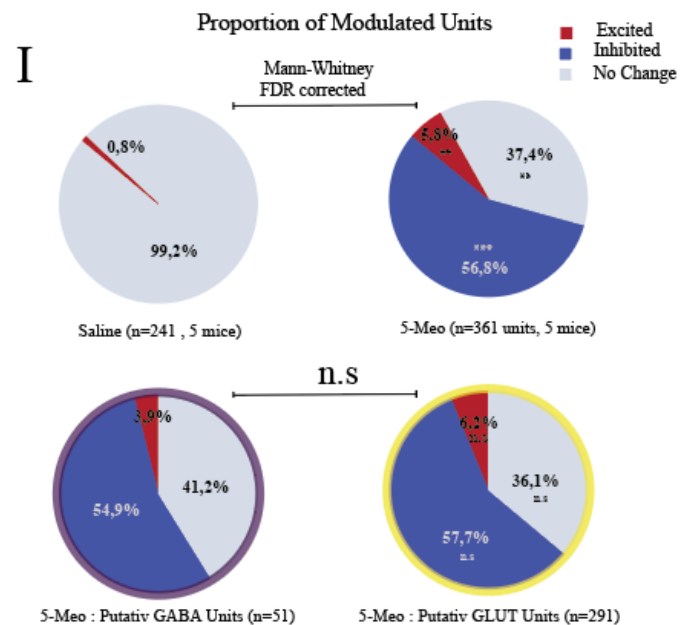
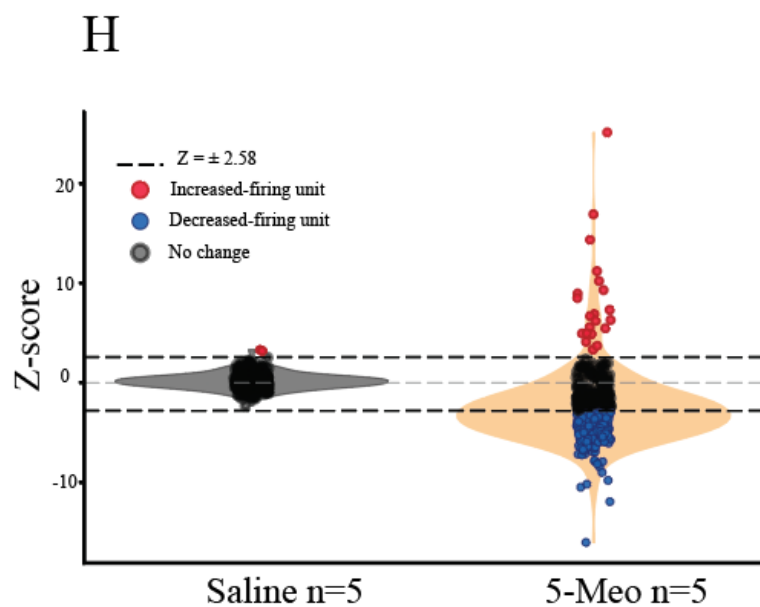
As shown in **Figure 5L**, both the **gamma power (black)** and the **mean Z-scored firing rate of excited units (red)** exhibit a remarkably similar temporal evolution following 5-MeO-DMT injection. Both signals show a **synchronous and sharp increase shortly after injection**, followed by a gradual and sustained decline. This visual parallel suggests a possible **co-modulation of gamma oscillations and excitatory neuronal firing** within the anterior insula.

While this alignment is striking, the relationship between the two signals **remains to be formally quantified**. Future analyses will assess whether this apparent synchrony reflects a true functional coupling—e.g., through spike–LFP phase-locking or cross-correlation—or whether both signals are independently modulated by a common upstream mechanism triggered by 5-MeO-DMT.



G





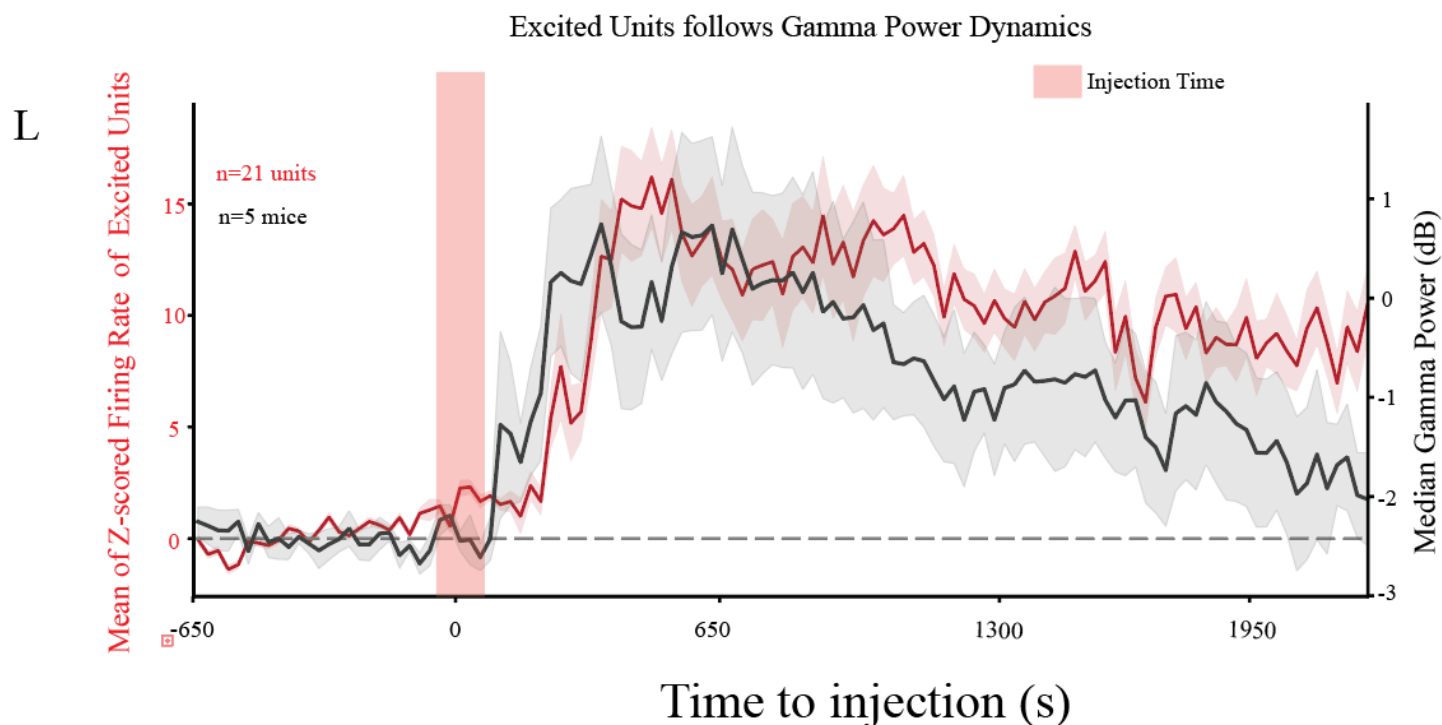


Figure 5. Single-unit classification and modulation of neuronal firing dynamics under 5-MeO-DMT.

A. Electrophysiological features extracted from average spike waveforms. Schematic representations of waveform metrics used to characterize single-unit activity. Parameters include peak-to-peak duration, repolarization slope, left peak angle, area under the curve (AUC), half-width duration, total waveform duration, peak amplitude asymmetry, and peak-to-trough ratio. These features were extracted from mean waveforms across channels per unit and used for unit classification and dimensionality reduction. **B. Principal component analysis (PCA) of waveform features across all recorded units.** Scatterplot of units ($n = 581$) in the space of the first three principal components derived from the waveform feature matrix. Each point represents one unit, positioned according to its scores on PC1 (30% variance explained), PC2 (23%), and PC3 (18%). PCA was applied to reduce dimensionality and visualize structure in the electrophysiological diversity of the recorded population. **C–D. K-means clustering of single units based on waveform features and PCA projections.** (C) Scatterplot of units ($n = 581$) in a 3D space defined by canonical waveform-derived features: half-width duration, peak-to-trough ratio, and mean firing rate. Units were clustered using k -means ($k = 2$), yielding two populations labeled as putative excitatory (black) and inhibitory (yellow) based on waveform geometry. (D) The same units were clustered in the space of the first three principal components (PC1–PC3) computed from full waveform metrics. Pie charts indicate the proportion of units assigned to each cluster in both approaches. Classification is based solely on unsupervised feature structure, without anatomical or optotagging confirmation. **E–F. Consistency between clustering methods for classifying single units.** (E) Confusion matrix comparing unit labels obtained via PCA-based clustering of waveform features versus physiologically defined categories (based on canonical waveform metrics). Each cell indicates the number of units classified into each combination of categories across all recorded units ($n = 581$). The overall agreement between methods is 85.2%, with 466 units classified as excitatory and 29 as inhibitory by both methods. (F) Agreement rate per animal between PCA- and physiology-based classifications. Each point represents one mouse; the vertical axis shows the percentage of units with matching classification labels across methods. Bars indicate mean $\pm$ standard deviation across mice ($85.6\% \pm 9.0\%$). Agreement values range from 69.2% to 100%, indicating consistency at both the group and individual levels. **G. Time-resolved firing rate Z-scores for all recorded units, aligned to injection.** Heatmaps represent normalized firing rates (Z-scores) for all units recorded in the saline (left, $n = 5$ mice, 241 units) and 5-MeO-DMT (right, $n = 6$ mice, 361 units) groups. Each row corresponds to one unit; time is aligned to injection ($t = 650$ s, dashed vertical line). Firing rates were computed in successive time bins and Z-scored relative to a pre-injection baseline. Units are ordered by their mean post-injection activity to visualize population-level changes. Color scale indicates Z-score range from -5 (blue) to $+5$ (red). **Figure H. Distribution of mean post-injection firing rate changes across single units.** Violin plots show the distribution of mean Z-scored firing rates (relative to baseline) for all units recorded in saline ($n = 5$ mice) and 5-MeO-DMT ($n = 5$ mice) groups. Each point represents one unit. Units with a post-injection Z-score $> +2.58$ (red) are classified as increased-firing; those with $Z < -2.58$ (blue) are classified as

decreased-firing; units within this range (gray) show no significant change. Dashed lines indicate $Z = \pm 2.58$ threshold corresponding to a two-tailed $p < 0.01$. Violin width reflects kernel density **I. Proportions of modulated units following 5-MeO-DMT administration.** Pie charts show the distribution of unit modulation types based on post-injection firing rate Z-scores: excited (red, $Z > +2.58$), inhibited (blue, $Z < -2.58$), and non-modulated (gray, $-2.58 < Z < +2.58$). Top panels: comparison of all units recorded in the saline group ($n = 241$ units, 5 mice) and 5-MeO-DMT group ($n = 361$ units, 5 mice). Bottom panels: modulation proportions in the 5-MeO-DMT group, separated by unit type based on waveform classification—putative GABAergic ($n = 51$) and putative glutamatergic units ($n = 291$). Significance of intergroup differences was tested using chi-squared or Mann–Whitney tests, as indicated. Asterisks denote levels of statistical significance ($p < 0.05$, $p < 0.01$, $p < 0.001$; n.s. = not significant). Percent values are indicated within each pie sector. **J–K. Temporal dynamics of firing rate modulation before and after 5-MeO-DMT injection.** (J) Mean Z-scored firing rates ($\pm$ SEM) are plotted over time for all units in the 5-MeO-DMT group ($n = 5$ mice, orange) and saline group ($n = 5$ mice, gray). Firing rates were computed in 10-second bins and normalized to each unit's pre-injection baseline. Time is aligned to injection ($t = 0$ s, red dashed line). The 5-MeO-DMT group shows a sustained reduction in population activity following injection. (K) Mean Z-scored firing rates ($\pm$ SEM) are shown separately for units classified as increased-firing (red) or decreased-firing (blue) within the 5-MeO-DMT group ($n = 5$ mice). Classification is based on post-injection Z-score thresholds (± 2.58). The plot illustrates opposite trajectories in excitatory and inhibitory unit responses following drug administration. **Figure L. Temporal alignment of gamma-band power and firing rate of excited units following 5-MeO-DMT injection.** Mean Z-scored firing rate (red, left y-axis) of excited units ($n = 21$ units from 5 mice) and median gamma-band power (30–80 Hz, gray, right y-axis, in dB) are plotted over time, aligned to the moment of injection ($t = 0$ s). Shaded regions indicate $\pm$ SEM across units (firing rate) and across animals (gamma power). Data are binned in 10-second intervals. A transient increase in both gamma power and firing rate is observed after injection, followed by gradual divergence. The shaded red bar marks the injection period.

GENERAL DISCUSSION

The present study reveals a striking and paradoxical reorganization of neuronal dynamics in the anterior insular cortex (aIC) following acute administration of 5-MeO-DMT. At first glance, the data suggest opposing processes: a robust suppression of firing rates across the majority of single units, and simultaneously, a pronounced increase in spectral power—most notably in the gamma range. Rather than representing analytical noise or artifact, this apparent contradiction points toward a genuine shift in the operational mode of cortical circuits under psychedelics, prompting a reevaluation of how excitation, inhibition, and synchrony co-exist and interact.

One of the most consistent findings was a decrease in alpha power and a concurrent emergence of gamma oscillations, both detected through conventional PSD methods and confirmed by FOOOF decomposition. These changes suggest a disruption in top-down control and thalamocortical coherence, alongside a rise in fast, locally coordinated rhythms. Gamma enhancement, especially when accompanied by structured peaks in the power spectrum, is often interpreted as a signature of active processing or binding. Yet in our data, this gamma increase unfolded in the context of global spiking inhibition—a seeming paradox that challenges simplistic interpretations of gamma as a direct proxy for excitatory drive.

At the unit level, while the majority of neurons were inhibited, a minority displayed significant increases in firing rate. This bimodal modulation raises the possibility that psychedelics do not uniformly dampen cortical output but instead reconfigure the contribution of specific subpopulations. One speculative but compelling hypothesis is that gamma oscillations emerge from precisely this selective recruitment of interneuron-rich microcircuits, which, despite reduced overall spiking, maintain—or even amplify—temporal coordination through inhibitory loops. In

this view, gamma is not a marker of excitation per se, but of rhythmic structure imposed by reorganized inhibitory networks.

This hypothesis gains support from spectral changes in the aperiodic component. The increase in FOOOF offset—reflecting a broadband elevation in signal power—may at first seem contradictory to reduced spiking. However, spectral power reflects not only action potentials but also subthreshold synaptic activity and population-level synchrony. Inhibitory processes, particularly those involving GABAergic interneurons, can generate strong rhythmic activity without driving spikes. Thus, what appears as “more activity” may reflect more organization, not more output.

This raises a deeper question: if spectral entropy increases under 5-MeO-DMT, does this imply an expansion of information, or merely a diversification of states? Entropy, in this context, may reflect neither noise nor chaos, but a redistribution of computational regimes—where previously dominant hierarchies dissolve and previously silent patterns emerge. Rather than disorder, the system may be entering a broader but structured dynamic repertoire. Notably, our findings in mice parallel those reported in human studies. Under DMT, alpha power decreases while gamma and broadband entropy increase (Timmermann et al., 2023). Similar dynamics have been reported for psilocybin and LSD, including increased brain energy, flattened hierarchical organization, and expanded dynamical repertoire (Atasoy et al., 2018; Vollenweider & Smallridge, 2022). The anterior insula, as a hub for interoception, salience, and conscious access, may be particularly sensitive to these reorganizations—both local and driven by interactions with other cortical or subcortical regions. The observed changes could be shaped by distal input, such as feedback from the prefrontal cortex or disinhibition from brainstem neuromodulatory systems, rather than being strictly locally generated.

Taken together, these findings do not simply confirm prior models—they complicate them. The data suggest that psychedelics like 5-MeO-DMT do not merely activate or inhibit brain regions, but fundamentally reorganize how activity is coordinated across scales. Suppression of firing does not preclude synchrony. Amplified rhythms can coexist with silenced populations. Gamma can be inhibitory. These are not contradictions to be resolved, but tensions to be explored. The anterior insula, with its central role in interoception, salience, and consciousness, may be especially susceptible to such reconfigurations.

In this light, the insula under 5-MeO-DMT may not be doing more or less, but doing differently—rebalancing its internal architecture to support alternate modes of processing. Whether this entails disengagement from sensorimotor mapping, heightened internal salience, or the breakdown of predictive coding hierarchies remains an open question. But the decoupling of spectral and unit-level activity provides a compelling lens through which to examine altered states of consciousness not as simple gain changes, but as reorganizations of the rules of engagement within cortical circuits.

Methodological Limitations

Despite the internal consistency of our results, several methodological constraints warrant caution. First, neuronal classification was based solely on extracellular waveform features, without ground-truth validation such as opto-tagging, introducing potential ambiguity—especially for intermediate

waveform types. Additionally, some statistical comparisons were performed at the unit level without correction for animal identity, raising concerns about pseudoreplication and inflated significance.

Temporal resolution was another limitation: firing rate modulations were averaged over long time windows, which may obscure transient or delayed effects. Likewise, the mechanisms underlying widespread inhibition remain unresolved—whether due to direct serotonergic modulation, interneuron activation, or broader network-level suppression.

At the spectral level, power analyses relied on a baseline normalization presumed stable across conditions, which may not hold uniformly. The use of 10-minute bins, while facilitating group-level comparisons, likely smoothed out fast, dynamic spectral changes. Furthermore, our analyses treated each frequency band independently, omitting cross-frequency interactions such as alpha–gamma antagonism or phase–amplitude coupling. Finally, no synchrony or connectivity metrics (e.g., spike–LFP coupling, population coherence) were employed, limiting our ability to assess intra-circuit reorganization.

Additionally, the modest sample size limits statistical power, especially for detecting fine-grained or interaction effects. Some analyses relied on linear mixed models, whose assumptions (e.g., homoscedasticity, normally distributed residuals) may not be fully met—particularly in higher frequency bands. Finally, although care was taken during preprocessing, potential non-neuronal contributions (e.g., movement artifacts) cannot be fully excluded and may have confounded some spectral measurements.

These limitations do not undermine the robustness of our findings but highlight the need for more temporally resolved, multivariate, and circuit-oriented approaches in future investigations.

Perspectives and Future Directions

Several questions raised by our findings point toward concrete avenues for future research. A central paradox lies in the coexistence of widespread unit-level inhibition with enhanced gamma-band synchrony. To resolve this, phase-locking analyses between spikes and LFPs—particularly in the gamma range—could determine whether inhibited or excited units remain temporally coupled to network rhythms, suggesting ongoing coordination despite reduced output.

To explore potential cross-frequency interactions, phase–amplitude coupling (PAC) analyses could reveal whether gamma oscillations are modulated by slower rhythms (e.g., delta–gamma or theta–gamma coupling), indicating a hierarchical reorganization of cortical dynamics. While spectral entropy can quantify the diversity and unpredictability of neural rhythms over time, complementary metrics applied to spike trains—such as spike train variability or information-theoretic complexity—may offer a distinct view of spiking structure. These tools could help determine whether the observed inhibition reflects disordered noise or a subtle reconfiguration of neural information flow.

Beyond these univariate measures, manifold learning techniques such as CEBRA -a contrastive embedding method for aligning high-dimensional neural data with behavioral or temporal structure

- provide a promising framework to uncover latent dynamical trajectories that conventional band-limited or time-averaged analyses cannot resolve. Applying CEBRA jointly to LFPs and spike trains could help identify shared or diverging neural trajectories across timescales and cell populations—for instance, transitions between drug-induced states, reoccurring population motifs, or low-dimensional attractors that escape traditional representations.

Addressing the ambiguity in unit classification will require more definitive ground-truth strategies. Ongoing and future efforts involving optogenetic tagging, cell-type-specific photometry, or in vitro patch-clamp recordings will be essential to validate waveform-based categorizations and disentangle inhibitory vs. excitatory contributions to the observed modulation. In parallel, time-resolved population analyses—such as peri-injection PSTHs, state trajectory clustering, or hidden Markov modeling—may reveal whether inhibition is global and sustained, or temporally structured and subtype-specific.

In parallel with these neural approaches, integrating behavioral data more explicitly could offer critical insights into the functional relevance of these dynamics. Specifically, linking ensemble neural activity to backward locomotion or immobility states may clarify whether the observed signatures reflect sensorimotor disruption, altered agency, or novel internal representations. Connecting brain dynamics to behaviorally relevant phenotypes will be key to interpreting the cognitive dimensions of the psychedelic state.

Together, these approaches promise to bridge the current gap between descriptive neurophysiology and mechanistic understanding. They may ultimately bring us closer to defining reliable biomarkers of psychedelic brain states—among which gamma oscillations in the anterior insula stand out as a compelling candidate.

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